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Supplementary Materials for

Momentum-exchange interactions in a Bragg atom interferometer suppress Doppler dephasing

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Supplementary materials

1 Atomic probe, Bragg and dressing laser

A single laser coupled into the cavity is used to realize the quantum non-demolition (QND) measurement of the momentum state populations, to create the dressing laser for momentum exchange interactions, and to realize Bragg rotations. We refer to this laser generically as the atomic probe, and we realize these various functions via rapid control of its frequency and phase as described below. In all cases, the relevant cavity resonance is stabilized to the blue of the atomic transition frequency ω_a with detuning about 500 MHz.

For the QND measurement we measure a time-averaged shift of the cavity resonance frequency that depends approximately linearly on the number of atoms in the $|F = 2, m_F = 0\rangle$ state. We do this by creating a probe tone incident on the cavity that is swept across resonance in time (12). We then detect this reflected light using homodyne detection. To achieve this, the atomic probe laser is blue detuned of the cavity by nominally 80 MHz. The homodyne reference beam or local oscillator is created by picking off a fraction of the laser light and shifting it to the red by 80 MHz using an acousto-optic modulator (AOM). The probe tone is generated using a fiber phase modulator weakly driven with 80 MHz. The lower phase modulation sideband serves as the actual tone used to measure the cavity frequency shift and is at the same frequency as (and phase coherent with) the homodyne reference laser. The much stronger carrier is nominally 80 MHz from the cavity resonance and therefore simply reflects off of the cavity and generates a heterodyne tone at 80 MHz in the homodyne detector. We detect the phase of this 80 MHz signal to monitor optical path length changes and then actively stabilize the homodyne detection quadrature via feeding back on the phase of the 80 MHz frequency sent to the AOM used to generate the homodyne reference beam. The cavity frequency shift is measured by sweeping the atomic probe laser's frequency over a range of 1 MHz from below to above the cavity resonance frequency in approximately 200 μs and fitting the observed dispersive signal from the homodyne detector.

Driving the Bragg rotations between $|p_0 - \hbar k\rangle$ and $|p_0 + \hbar k\rangle$ requires two different optical tones on the atomic probe beam separated by ω_z. We realize this by first red shift the atomic probe by 75 MHz with one AOM and then blue shift it back with another AOM driven with two radio frequency (RF) tones at 75 MHz and $(75 - \omega_z/2\pi)$ MHz before sending it into the fiber phase modulator. Because the atoms accelerate under gravity, we linearly chirp the frequency separation $\omega_z/2\pi$ between the two sidebands by 25.11 kHz/ms to compensate the changing Doppler shift (12). This light is then sent through the same fiber phase modulator as above before being injected into the cavity. By stabilizing the atomic probe carrier at 77 MHz from the cavity, the lower phase modulation sidebands now consists of two tones separated by ω_z that are 3 MHz detuned from the cavity resonance. These tones non-resonantly enter the cavity and drive the Bragg rotations. With the detuning $3\ \mathrm{MHz} \gg \kappa$, we suppress any frequency noise to amplitude or phase noise conversion that would degrade the fidelity of the Bragg rotations. For all the Bragg pulses applied in the interferometer sequence, the Rabi frequency is 8.3 kHz, giving a π-pulse duration of $60.\ \mu$s.

For driving the momentum exchange interaction, we apply only one RF tone at 75 MHz on the (blue shifting) AOM before the phase modulation as for the QND measurement and fast switch the RF frequency for driving the fiber phase modulator to put the atomic probe tone about few hundreds kHz detuned from the dressed resonance. For the short durations for which we applied the dressing laser, it was not necessary to chirp the frequency of the dressing laser to hold the interaction strength approximately constant.

2 Initial state preparation and readout

The atoms are initially prepared via velocity-dependent Raman transitions in a very narrow momentum range and hyperfine state given by$|F = 2, m_F = 0, p_0 - \hbar k\rangle$. This specific hyperfine state is favorable for the velocity-selection since using the Rb clock states avoids broadening of the transition due to magnetic fields. However, to exploit the more favorable Clebsch-Gordan coefficients of the $|F = 2, m_F = 2\rangle$ to excited $|F' = 3, m_F = 3\rangle$ cycling transition, we use a series of microwave π-pulses to transfer the atoms into $|F = 2, m_F = 2\rangle$ without significantly shifting their momentum states. We first apply a microwave π-pulse to transfer atoms into $|F = 1, mF = 1, p_0 - \hbar k\rangle$, blow away any untransferred atoms with a laser beam on resonance with $|F = 2\rangle \rightarrow |F' = 3\rangle$ transition then a second microwave π-pulse transfer the atoms into $|F = 2, m_F = 2, p_0 - \hbar k\rangle$. This configuration is the initial state from which we realize the Bragg interferometer and momentum exchange interactions.

To read out the populations in individual momentum states at the end of the experimental sequences, we begin by using microwave π-pulses to map the population back from $|F = 2, m_F = 2\rangle$ to $|F = 1, m_F = 0\rangle$. The process is the reverse of the above, but with an added π-pulse at the end to transfer atoms from $|F = 2, m_F = 0\rangle$ to $|F = 1, m_F = 0\rangle$. We then apply a Raman π-pulse (two-photon Rabi frequency $4.2\,\text{kHz}$) for transferring atoms from $|F = 1, m_F = 0, p_0 - \hbar k\rangle$ to $|F = 2, m_F = 0, p + \hbar k\rangle$ (12). A QND measurement is then used to measure the number of atoms or population as described above. We then blow away the $|F = 2\rangle$ atoms with a laser beam resonant with the atomic $|F = 2 \rightarrow 3'\rangle$ transition. We then measure the population in $p_0 + \hbar k$ by applying a Raman π-pulse again with the appropriate two-photon detuning and perform a second QND measurement. The momentum exchange interaction's residual superradiance may transfer atoms into adjacent momentum states $p_0 \pm 3\hbar k$. We iterate the above procedure to measure these populations as well. Thus, on every run of the experiment, we measure the populations in the four momentum states $p_0 \pm \hbar k$ and $p_0 \pm 3\hbar k$.

3 Derivation of momentum exchange Hamiltonian

3.1 Original Hamiltonian in momentum basis

Here we start with the full atom-cavity Hamiltonian in the lab frame given as $\hat{H}_{\text{lab}} = \hat{H}_{\text{atom}} + \hat{H}_{\text{light}} + \hat{H}_{\text{int}}$ with the following second quantized form:

$$\hat{H}_{\text{light}} = \hbar\sqrt{\kappa_1}\left(\alpha_d \hat{a}^\dagger e^{-i\omega_d t} + \alpha_d^* \hat{a} e^{i\omega_d t}\right) + \hbar\omega_c \hat{a}^\dagger \hat{a}, \tag{S1}$$

$$\hat{H}_{\text{atom}} = \sum_{\beta=g,e} \int \hat{\psi}_\beta^\dagger(Z) \frac{\hat{p}^2}{2m} \hat{\psi}_\beta(Z)\, dZ + \hbar\omega_a \int \hat{\psi}_e^\dagger(Z)\hat{\psi}_e(Z)\, dZ, \tag{S2}$$

$$\hat{H}_{\text{int}} = \hbar g_0 \int \cos\left(kZ\right) \left[\hat{a}\hat{\psi}_e^\dagger(Z)\hat{\psi}_g(Z) + \hat{a}^\dagger \hat{\psi}_g^\dagger(Z)\hat{\psi}_e(Z)\right] dZ. \tag{S3}$$

Here, α_d is the complex amplitude of the incident dressing laser with photon flux $|\alpha_d|^2$ in unit of photons per second, and $\kappa_1 = T_1/\tau_{\text{rnd}}$ is the input coupling of the cavity with T_1 the power transmission coefficient for the input mirror and τ_{rnd} the optical round trip time in the cavity. $\hat{\psi}_{e(g)}^\dagger(Z)$ are the field creation operator for an atom at position Z in excited state $|e\rangle \equiv |F = 3, m_F = 3\rangle$ and ground state $|g\rangle \equiv |F = 2, m_F = 2\rangle$, ω_a is the energy splitting between ground and excited state, $\delta_d = \omega_d - \omega_c$ is the detuning of the dressing laser from the empty cavity, and $\Delta_a = \omega_d - \omega_a$ is the detuning of the dressing laser from the atomic transition frequency. We rewrite the Hamiltonian in the rotating frame of the dressing laser at ω_d, with the Hamiltonian used to construct the appropriate unitary $\hat{U} = \exp\left(i\hat{H}_r t\right)$ given by $\hat{H}_r = \hbar\omega_d \hat{a}^\dagger \hat{a} + \hbar\omega_d \int \hat{\psi}_e^\dagger(Z)\hat{\psi}_e(Z)\, dZ$. In this frame, the Hamiltonian $\hat{H}_d = \hat{U}^\dagger \hat{H}_{\text{lab}} \hat{U} + i\hbar \frac{\partial \hat{U}^\dagger}{\partial t}\hat{U}$ takes the form:

$$\begin{aligned}\hat{H}_d = {} & \hbar\sqrt{\kappa_1}\left(\alpha_d \hat{a}^\dagger + \alpha_d^* \hat{a}\right) - \hbar\delta_d \hat{a}^\dagger \hat{a} - \hbar\Delta_a \int \hat{\psi}_e^\dagger(Z)\hat{\psi}_e(Z)\, dZ + \sum_{\tau=g,e} \int \hat{\psi}_\tau^\dagger(Z)\frac{\hat{p}^2}{2m}\hat{\psi}_\tau(Z)\, dZ \\ & + \hbar g_0 \int \cos(kZ) \left[\hat{a}\hat{\psi}_e^\dagger(Z)\hat{\psi}_g(Z) + \hat{a}^\dagger\hat{\psi}_g^\dagger(Z)\hat{\psi}_e(Z)\right] dZ.\end{aligned} \tag{S4}$$

Furthermore, we assume the excited state population and atomic spontaneous emission are negligible during the relevant time scales with $\Delta_a \gg g_0\sqrt{\langle \hat{a}^\dagger \hat{a}\rangle}$ and $\Delta_a \gg \gamma$, where γ is the spontaneous population decay rate from the optically excited state. In this limit, we can adiabatically eliminate the excited state $|e\rangle$, which leads to the effective Hamiltonian describing dispersive coupling between atoms in the ground state manifold and cavity:

$$
\begin{aligned}
\hat{H}_{\text{disp}} &= \int \hat{\psi}^\dagger(Z)\left[\frac{\hat{p}^2}{2m} + \frac{\hbar g_0^2}{\Delta_a}\cos^2(kZ)\hat{a}^\dagger\hat{a}\right]\hat{\psi}(Z)\,dZ - \hbar\delta_d\hat{a}^\dagger\hat{a} + \hbar\sqrt{\kappa_1}\left(\alpha_d\hat{a}^\dagger + \alpha_d^*\hat{a}\right) \\
&= \int \hat{\psi}^\dagger(Z)\left[\frac{\hat{p}^2}{2m} + \frac{\hbar g_0^2}{4\Delta_a}\left(e^{i2kZ} + e^{-i2kZ}\right)\hat{a}^\dagger\hat{a}\right]\hat{\psi}(Z)\,dZ \\
&\quad - \hbar\Delta_d\hat{a}^\dagger\hat{a} + \hbar\sqrt{\kappa_1}\left(\alpha_d\hat{a}^\dagger + \alpha_d^*\hat{a}\right),
\end{aligned}
\tag{S5}
$$

with $\hat{\psi}(Z) \equiv \hat{\psi}_g(Z)$. This dispersive Hamiltonian tells us that an atom created at an anti-node of the cavity (i.e. $\cos^2(kZ) = 1$) causing a shift of the cavity resonance by $\frac{g_0^2}{\Delta_a}$ as stated in the main text. For a uniform distribution of atoms along the cavity axis, the cavity's dressed resonance frequency is on-average shifted by $\frac{Ng_0^2}{2\Delta_a}$. The detuning $\Delta_d = \delta_d - \frac{Ng_0^2}{2\Delta_a}$ can be physically interpreted then as the detuning of the dressing laser from the average dressed cavity resonance frequency.

We then perform a Fourier transformation on the atomic field operator $\hat{\psi}(Z)$ to the momentum space as $\hat{\psi}(Z) = \int_{-\infty}^{\infty}\hat{\psi}(p)e^{ipZ/\hbar}\,dp$, here $\hat{\psi}(p)$ annihilates an atom with momentum p. Due to the narrow momentum distribution after velocity selection with rms momentum spread $\sigma_p \ll \hbar k$, we only focus on momentum states within the range $[p_0 - 2\hbar k, p_0 + 2\hbar k]$. We further define $\hat{\psi}_\uparrow(p) \equiv \hat{\psi}(p + p_0 + \hbar k), \quad \hat{\psi}_\downarrow(p) \equiv \hat{\psi}(p + p_0 - \hbar k)$, with $p \in [-\hbar k, \hbar k]$. The operator $\hat{\psi}_\uparrow(p)$ annihilates an atom with momentum $p + p_0 + \hbar k_c$, while $\hat{\psi}_\downarrow(p)$ annihilates an atom with momentum $p + p_0 - \hbar k$.

The original atomic field operator can now be expressed as

$$
\begin{aligned}
\hat{\psi}(Z) &\approx \int_{p_0-2\hbar k}^{p_0}\hat{\psi}(p)e^{ipZ/\hbar}\,dp + \int_{p_0}^{p_0+2\hbar k}\hat{\psi}(p)e^{ipZ/\hbar}\,dp \\
&= e^{ip_0Z/\hbar}\int_{-\hbar k}^{+\hbar k} e^{ipZ/\hbar}\left(\hat{\psi}_\downarrow(p)e^{-ikZ} + \hat{\psi}_\uparrow(p)e^{ikZ}\right)\,dp.
\end{aligned}
\tag{S6}
$$

Substituting the approximate Fourier transform to momentum space into the dispersive Hamiltonian Eq. (S5), we find

$$
\begin{aligned}
\hat{H}_{\text{disp}} =& \int \hat{\psi}^\dagger(Z) \left[\frac{\hat{p}^2}{2m} + \frac{g_0^2}{4\Delta_a}(e^{2ikZ} + e^{-2ikZ})\hat{a}^\dagger\hat{a} \right] \hat{\psi}(Z)\, dZ + \hat{H}_{\text{cavity}} \\
\approx& \int_{-\hbar k}^{+\hbar k} \left[\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p)\frac{(p+p_0-\hbar k)^2}{2m} + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p)\frac{(p+p_0+\hbar k)^2}{2m} \right] dp + \hat{H}_{\text{cavity}} \\
&+ \frac{g_0^2}{4\Delta_a}\hat{a}^\dagger\hat{a} \int dZ \int_{-\hbar k}^{+\hbar k}\int_{-\hbar k}^{+\hbar k} \left[\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(q)e^{i(q-p)Z} + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q)e^{i(p-q)Z} \right] dp\, dq \\
=& \frac{1}{2}\int_{-\hbar k}^{+\hbar k} \left[\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p) - \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p) \right] \left(\omega_z + \frac{2\hbar k p}{m} \right) dp + \hat{H}_{\text{cavity}} \\
&+ \frac{g_0^2}{4\Delta_a}\hat{a}^\dagger\hat{a} \int_{-\hbar k}^{+\hbar k}\int_{-\hbar k}^{+\hbar k} \delta(p-q) \left[\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(q) + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q) \right] dp\, dq,
\end{aligned}
\tag{S7}
$$

where the delta function that naturally emerges above arises from the physical fact that the two-photon processes can only couple momentum states separated by two photon momentum recoils $2\hbar k$.

We can break the above dispersive Hamiltonian into physically meaningful terms as

$$
\hat{H}_{\text{disp}} = \hat{H}_{\text{z}} + \hat{H}_{\text{in}} + \hat{H}_{\text{cavity}} + \hat{H}_{\text{atom-cavity}}. \tag{S8}
$$

The average kinetic energy difference $\hbar\omega_z = 2\hbar k p_0/m$ between the two momentum states $p_0 + p \pm \hbar k$ (i.e. independent of p) is given by

$$
\hat{H}_{\text{z}} = \frac{\hbar\omega_z}{2}\int_{-\hbar k}^{+\hbar k} \left[\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p) - \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p) \right] dp. \tag{S9}
$$

The inhomogeneous kinetic energy difference between the two momentum states $p_0 + p \pm \hbar k$ (i.e. dependent on p) is captured by

$$
\hat{H}_{\text{in}} = \frac{\hbar k}{m}\int_{-\hbar k}^{+\hbar k} p \left[\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p) - \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p) \right] dp. \tag{S10}
$$

The driven cavity is described by

$$
\hat{H}_{\text{cavity}} = -\hbar\Delta_d\hat{a}^\dagger\hat{a} + \hbar\sqrt{\kappa_1}\left(\alpha_d\hat{a}^\dagger + \alpha_d^*\hat{a} \right). \tag{S11}
$$

The remaining atom-cavity coupling is described by

$$\hat{H}_{\mathrm{atom-cavity}} = \hat{a}^\dagger \hat{a}\, \frac{\hbar g_0^2}{4\Delta_a} \int_{-\hbar k}^{+\hbar k} \left[\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(p) + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p)\right]\, dp. \tag{S12}$$

In the following, we will describe how to map the above Hamiltonian to a pseudo-spin model and show how to eliminate the cavity field and arrive at an effective atom-only interaction.

3.2 Mapping from momentum states to pseudo-spins

Using a Schwinger-boson representation, we can define pseudo-spin operators based on the two momentum states at $p + p_0 \pm \hbar k$. First we define ladder operators

$$\hat{j}_+ = \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p), \quad \hat{j}_- = \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(p), \tag{S13}$$

from which we can also construct the spin angular momentum operators

$$\begin{aligned} \hat{j}_x(p) &= \frac{1}{2}\left[\hat{j}_+(p) + \hat{j}_-(p)\right], \\ \hat{j}_y(p) &= \frac{1}{2i}\left[\hat{j}_+(p) - \hat{j}_-(p)\right], \\ \hat{j}_z(p) &= \frac{1}{2}\left[\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p) - \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p)\right], \end{aligned} \tag{S14}$$

where the operators satisfy the commutation relation $\left[\hat{j}_x(p), \hat{j}_y(p)\right] = i\epsilon_{xyz}\,\hat{j}_z(p)$ plus all permutations of x, y, z, and where ϵ_{xyz} is the Levi-Civita symbol. Summing over all momentum states, we then define the collective spin operators

$$\hat{J}_\alpha = \int_{-\hbar k}^{\hbar k} \hat{j}(p)\, dp, \quad \alpha \in [x, y, z, +, -]. \tag{S15}$$

Using only spin operators, we can now express the dispersive Hamiltonian Eq. (S8) in the pseudo-spin basis

$$\hat{H}_{\mathrm{disp}} = \hbar\omega_z \hat{J}_z + \frac{\hbar g_0^2}{4\Delta_a}\hat{a}^\dagger\hat{a}(\hat{J}_+ + \hat{J}_-) - \hbar\Delta_d\hat{a}^\dagger\hat{a} + \hbar\sqrt{\kappa_1}\left(\alpha_d\hat{a}^\dagger + \alpha_d^*\hat{a}\right) + \hat{H}_{\mathrm{in}}. \tag{S16}$$

In this pseudo-spin basis, we also re-write the inhomogeneous Hamiltonian as

$$\hat{H}_{\rm in} = \frac{2\hbar k}{m} \int_{-\hbar k}^{\hbar k} p\, \hat{j}_z(p)\, dp. \tag{S17}$$

In this pseudo-spin picture, the inhomogeneous kinetic energy can be thought of as an effective inhomogeneous magnetic field that shifts the energy difference between spin up and spin down with a linear dependence on p.

The time evolution of the density matrix describing the atom-cavity coupled system is given by the following Master equation:

$$\frac{d}{dt}\rho = -\frac{i}{\hbar}[\hat{H}_{\rm disp}, \rho] + \hat{L}\rho\hat{L}^\dagger - \frac{1}{2}\{\hat{L}^\dagger\hat{L}, \rho\}, \tag{S18}$$

where the jump operator describing the cavity power decay at rate κ is given by

$$\hat{L} = \sqrt{\kappa}\hat{a}. \tag{S19}$$

3.3 Effective Hamiltonian from second-order perturbation theory

In order to reach an effective atom-atom interaction from the Hamiltonian Eq. (S16), we eliminate the cavity field with second-order perturbation theory. First, we express the cavity operator $\hat{a}$ as a cavity operator $\hat{b}$ that captures the cavity field dynamics around an average field described by a time-independent complex number

$$\hat{a} = \alpha_0 + \hat{b}, \tag{S20}$$

$$\alpha_0 = \frac{\alpha_d\sqrt{\kappa_1}}{\Delta_d + i\kappa/2}, \tag{S21}$$

where the complex number α_0 in unit of $\sqrt{\rm photons}$ is the dressing laser's field amplitude that builds up inside the cavity at a dressing laser detuning Δ_d. Substituting Eq. (S20) back into the dispersive Hamiltonian Eq. (S16), the Hamiltonian and jump operators become

$$
\begin{aligned}
\hat{H}_{\alpha_0} &= \hat{H}_0 + \hat{V}, \\
\hat{H}_0 &= \hbar\omega_z \hat{J}_z - \hbar\Delta_d \hat{b}^\dagger \hat{b}, \\
\hat{V} &= \frac{\hbar g_0^2}{4\Delta_a}\left(\hat{J}_+ + \hat{J}_-\right)\left(\alpha_0 \hat{b}^\dagger + \alpha_0^* \hat{b}\right), \\
\hat{L} &= \sqrt{\kappa}\hat{b}.
\end{aligned} \tag{S22}
$$

Importantly, the interaction $\hat{V}$ can be interpreted as a field displacement operator acting on the cavity, but the amplitude of the displacement depends on the atomic degree of freedom $\hat{J}_+ + \hat{J}_-$. The term $\hat{J}_+ + \hat{J}_-$ will oscillate at frequency ω_z due to the $\hbar\omega_z\hat{J}_z$ term in $\hat{H}_0$. This term leads to the generation of modulation sidebands on the dressing laser light inside the cavity at absolute optical frequency $\omega_d \pm \omega_z$. In this picture, this is the origin of the modulation sidebands discussed in the main text. Conceptually, the sidebands that are generated represent the first-order perturbative modification of the cavity field about the zeroth order response α_0.

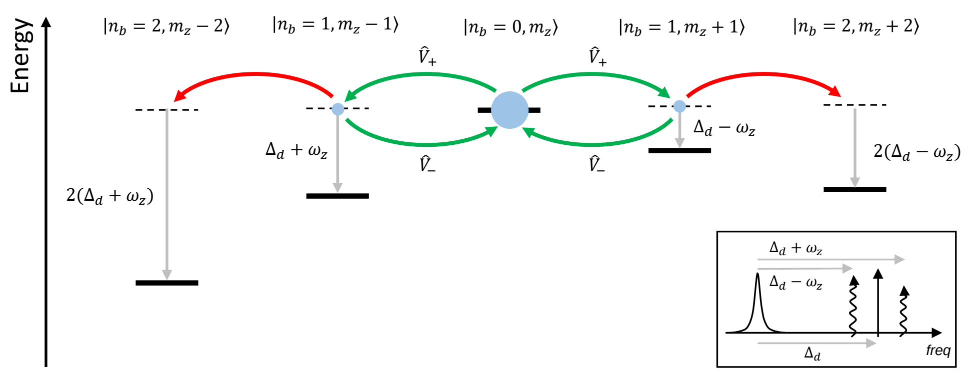

Figure S1: **Second order perturbation theory.** Relative energy level diagrams in the rotating frame of the dressing laser when applied at detuning Δ_d from the dressed cavity. In the second order perturbation theory, the initial states are coupled to the singly excited states and back with the processes labeled as green arrows. Due the the large energy offset $2(\Delta_d \pm \omega_z)$, the processes coupling doubly excited states (red arrows) are energetically forbidden.

To arrive at an atom-only Hamiltonian, we now go to the second order in perturbation theory by assuming that the $\left|\left\langle \hat{b} \right\rangle\right| \ll 1$ or equivalently the total amplitude of the generated sidebands is small, which is true if $|\Delta_d \pm \omega_z| \gg \sqrt{N}|\alpha_0|g_0^2/4\Delta_a$. In this limit, we can use $\hat{V}$ as a perturbation on $\hat{H}_0$ and adiabatically eliminate the photon excitation to obtain an effective Hamiltonian in the zero photon subspace. We consider the basis $|n_b, m_z\rangle$, where n_b is the photon number such that $\hat{b}^\dagger\hat{b}\,|n_b, m_z\rangle = n_b\,|n_b, m_z\rangle$ and m_z is the z-projection of the collective operator $\hat{J}_z$ such that $\hat{J}_z\,|n_b, m_z\rangle = m_z\,|n_b, m_z\rangle$. Consider an initial state of the system $|\text{initial}\rangle = |n_b = 0, m_z\rangle$. The only intermediate states $|\text{intermediate}\rangle$ such that $|\langle\text{intermediate}|\,\hat{V}\,|\text{initial}\rangle|^2 \neq 0$ are $|\text{intermediate}\rangle = |n_b = 1, m_z \pm 1\rangle$. We will see such terms give rise to the exchange interactions $\hat{J}_+\hat{J}_-$ and $\hat{J}_-\hat{J}_+$. At the same order in perturbation theory, there are additional non-zero matrix elements of the form $\langle n_b = 2, m_z \pm 2|\,\hat{V}\,|n_b = 1, m_z \pm 1\rangle\,\langle n_b = 1, m_z \pm 1|\,\hat{V}\,|n_b = 0, m_z\rangle \neq 0$. Such terms give rise to pair raising $\hat{J}_+^2$ and pair lowering $\hat{J}_-^2$ interactions (39,70). However these terms can be neglected since the initial and final states differ by $2\hbar\,(\Delta_d \pm \omega_z)$ in energy. After identifying the intermediate states, we can re-write the Hamiltonian $\hat{H}_0$ in the $|n_b, m_z\rangle$ basis as

$$\begin{aligned}\hat{H}_0 = &\sum_{m_z}(\omega_z - \Delta_d)\,|1, m_z+1\rangle\,\langle 1, m_z+1| - (\omega_z + \Delta_d)\,|1, m_z-1\rangle\,\langle 1, m_z-1| \\ &+ \sum_{m_z} m_z\omega_z\,(|1, m_z+1\rangle\,\langle 1, m_z+1| + |0, m_z\rangle\,\langle 0, m_z| + |1, m_z-1\rangle\,\langle 1, m_z-1|)\,,\end{aligned} \tag{S23}$$

where the second line can be ignored as a constant for the perturbation theory. And the $\hat{V} = \hat{V}_- + \hat{V}_+$ with

$$\begin{aligned}\hat{V}_+ = \sum_{m_z}\frac{g_0^2}{2\Delta_a}\alpha_0\Bigg[&\sqrt{\frac{N}{2}(\frac{N}{2}+1) - m_z(m_z+1)}\,|1, m_z+1\rangle\,\langle 0, m_z| \\ &+ \sqrt{\frac{N}{2}(\frac{N}{2}+1) - m_z(m_z-1)}\,|1, m_z-1\rangle\,\langle 0, m_z|\Bigg]\end{aligned} \tag{S24}$$

where $\hat{V}_- = \hat{V}_+^\dagger$. Second-order perturbation theory including the damping gives rise to an non-Hermitian Hamiltonian

$$\begin{aligned}\hat{H}_{\mathrm{nh}} =&\hat{H}_0 - \frac{i}{2}\hat{L}^\dagger \hat{L}\\ =&\sum_{m_z}\Big[(\omega_z - \Delta_d - i\kappa/2)\left|1, m_z+1\right\rangle\left\langle 1, m_z+1\right| \\ &-(\omega_z + \Delta_d + i\kappa/2)\left|1, m_z-1\right\rangle\left\langle 1, m_z-1\right|\Big],\end{aligned} \tag{S25}$$

from which we derive the effective Hamiltonian

$$\begin{aligned}\hat{H}_{\mathrm{eff}} &= -\frac{1}{2}\hat{V}^-\left[\hat{H}_{\mathrm{nh}}^{-1} + \left(\hat{H}_{\mathrm{nh}}^{-1}\right)^\dagger\right]\hat{V}^+\\ &= \chi_+\hat{J}_+\hat{J}_- + \chi_-\hat{J}_-\hat{J}_+ = \underbrace{(\chi_+ + \chi_-)}_{\chi}(\hat{J}^2 - \hat{J}_z^2) + (\chi_+ - \chi_-)\hat{J}_z,\end{aligned} \tag{S26}$$

with the interaction strengths

$$\begin{aligned}\chi_+ &= \left(\frac{g_0^2}{4\Delta_a}\right)^2 \frac{|\alpha_d|^2\kappa_1}{\Delta_d^2 + \kappa^2/4}\frac{\Delta_d + \omega_z}{(\Delta_d + \omega_z)^2 + \kappa^2/4},\\ \chi_- &= \left(\frac{g_0^2}{4\Delta_a}\right)^2 \frac{|\alpha_d|^2\kappa_1}{\Delta_d^2 + \kappa^2/4}\frac{\Delta_d - \omega_z}{(\Delta_d - \omega_z)^2 + \kappa^2/4}.\end{aligned} \tag{S27}$$

The effective jump operator describing the dissipation process is found as

$$\begin{aligned}\hat{L}_{\mathrm{eff}} =&\hat{L}\hat{H}_{\mathrm{nh}}^{-1}\hat{V}_+\\ =&\sum_{m_z}\frac{g_0^2\alpha_0\sqrt{\kappa}}{4\Delta_a}\left(\frac{\sqrt{\frac{N}{2}(\frac{N}{2}+1) - m(m+1)}}{\omega_z - \Delta_d - i\kappa/2}\left|0, m_z+1\right\rangle\left\langle 0, m_z\right|\right.\\ &\left. + \frac{\sqrt{\frac{N}{2}(\frac{N}{2}+1) - m(m-1)}}{-\omega_z - \Delta_d - i\kappa/2}\left|0, m_z-1\right\rangle\left\langle 0, m_z\right|\right)\\ =&\frac{g_0^2\alpha_0\sqrt{\kappa}}{4\Delta_a}\left(\frac{1}{\omega_z - \Delta_d - i\kappa/2}\hat{J}_+ + \frac{1}{-\omega_z - \Delta_d - i\kappa/2}\hat{J}_-\right),\end{aligned} \tag{S28}$$

which corresponds to the two superradiance processes where the dressing laser is $\pm\omega_z$ detuned from the dressed cavity resonance and the atoms change their momentum states with photons from the modulation sideband escaping from the cavity.

Given the effective Hamiltonian and jump operators above, we can calculate Heisenberg equations of motion for the spin operators under mean-field approximation:

$$\begin{aligned}
\left\langle \dot{\hat{J}}_+ \right\rangle &= i\omega_z \left\langle \hat{J}_+ \right\rangle - i\left[2(\chi_+ + \chi_-) + i(\Gamma_+ - \Gamma_-)\right] \left\langle \hat{J}_+ \right\rangle \left\langle \hat{J}_z \right\rangle, \\
\left\langle \dot{\hat{J}}_- \right\rangle &= -i\omega_z \left\langle \hat{J}_- \right\rangle + i\left[2(\chi_+ + \chi_-) - i(\Gamma_+ - \Gamma_-)\right] \left\langle \hat{J}_- \right\rangle \left\langle \hat{J}_z \right\rangle, \\
\left\langle \dot{\hat{J}}_z \right\rangle &= -(\Gamma_+ - \Gamma_-) \left\langle \hat{J}_+ \right\rangle \left\langle \hat{J}_- \right\rangle,
\end{aligned} \tag{S29}$$

with the superradiance or collective emission rate

$$\Gamma_\pm = |\alpha_0|^2 \left(\frac{g_0^2}{4\Delta_a}\right)^2 \frac{\kappa}{(\Delta_d \pm \omega_z)^2 + \kappa^2/4}. \tag{S30}$$

In the relevant limit for momentum-exchange $|\Delta_d \pm \omega_z| \gg \kappa$,

$$\Gamma_\pm \approx |\alpha_0|^2 \left(\frac{g_0^2}{4\Delta_a}\right)^2 \frac{\kappa}{(\Delta_d \pm \omega_z)^2}, \tag{S31}$$

and

$$\chi_\pm \approx |\alpha_0|^2 \left(\frac{g_0^2}{4\Delta_a}\right)^2 \frac{1}{\Delta_d \pm \omega_z}. \tag{S32}$$

From this, one sees that the ratio between the collective emission rate (undesired) to interaction rate (desired) is $\Gamma_\pm/\chi_\pm = \frac{\kappa}{\Delta_d \pm \omega_z}$. Therefore, it is desirable to work with higher cavity finesse and thus smaller linewidth κ.

3.4 Impact from adjacent momentum states

In the previous sections, we focus on an effective two-level system with two different momentum states, here denoted by p_0 and p_1 for simplicity. To study the impact from other momentum states outside the original two-level system, we now include more momentum states centered at $p_m = p_0 + 2m\hbar k$ with $\hat{a}_m^\dagger$ defined as operators that create an atom in the momentum state p_m. Using the Schwinger-boson representation, we then define

$$\hat{S}_m^+ = \hat{a}_{m+1}^\dagger \hat{a}_m, \quad \hat{S}_m^z = \frac{\hat{a}_{m+1}^\dagger \hat{a}_{m+1} - \hat{a}_m^\dagger \hat{a}_m}{2}. \tag{S33}$$

We now consider the cavity mediated interactions taking the general form of $\chi^+_{mn}\hat{S}^+_m\hat{S}^-_n + \chi^-_{mn}\hat{S}^-_n\hat{S}^+_m$. We first consider the exchange processes with $m = n = -1, 2$ in Fig. S2A that connect the original two-level system with the neighboring momentum states. These processes only occur if there are atoms in states p_{-1} and p_2 with corresponding populations $N_{-1} = \langle \hat{a}^\dagger_{-1}\hat{a}_{-1}\rangle$ and $N_2 = \langle \hat{a}^\dagger_2\hat{a}_2\rangle$. We find that these neighboring states contribute to the original Hamiltonian with extra terms $\hat{H}_N \approx \chi N_n \left(\hat{a}^\dagger_0\hat{a}_0 + \hat{a}^\dagger_1\hat{a}_1\right) + \Delta N_n \hat{S}^z_0$ with $N_n = N_2 + N_{-1}$ being the total population and $\Delta N_n = N_2 - N_{-1}$ the difference in population for the two neighboring states. Thus we see that neither the OAT dynamics or the energy gap protection is affected by these processes.

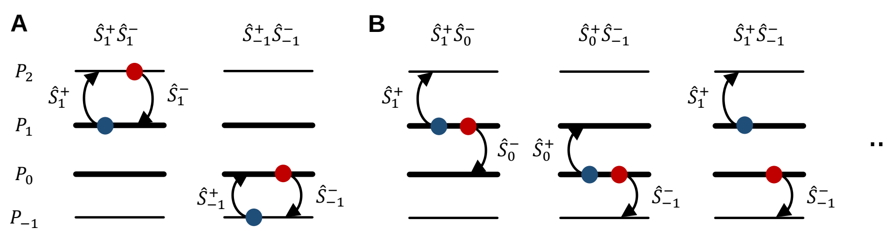

Figure S2: **Interaction $\hat{S}^+_m\hat{S}^-_n$ involving other momentum states.** **(A)** Exchange processes ($m = n$) involving other momentum states, which we neglect with the absence of populations in p_{-1} and p_2 momentum states. **(B)** Pair production processes ($m \neq n$) involving other momentum states, which represents the processes that annihilate a pair of atoms and subsequently create a pair in two other momentum states.

We next consider the pair production processes shown in Fig. S2B with $m \neq n$. These processes do not conserve energy by the amount $\omega_{mn} = \omega_m - \omega_n = 8\,(m-n)\,\omega_{rec}$, where the transition frequency between p_m and p_{m+1} is $\omega_m = \omega_z + 8m\omega_{rec}$ and $\omega_{rec} = \hbar k^2/2m \approx 2\pi \times 3.77$ kHz is the recoil frequency. With respect to the rotating frame defined by ω_z, the time-dependent pair production processes take the form of

$$\begin{aligned} \tilde{H}_P &= \sum_{m>n} (\chi^+_{mn}\hat{S}^+_m\hat{S}^-_n + \chi^-_{mn}\hat{S}^-_n\hat{S}^+_m)e^{-i\omega_{mn}t} + h.c. \\ &= \sum_{m>n} \chi_{mn}\hat{S}^+_m\hat{S}^-_n e^{-i\omega_{mn}t} + h.c. \end{aligned} \tag{S34}$$

with $\chi_{mn} = \chi^+_{mn} + \chi^-_{mn}$. Here we use the property $[\hat{S}^+_m, \hat{S}^-_n] = [\hat{a}^\dagger_{m+1}\hat{a}_m, \hat{a}^\dagger_n\hat{a}_{n+1}] = 0$ for $m \neq n$.

We further define $\hat{h}_{mn} = \chi_{mn}\hat{S}^+_m\hat{S}^-_n$, which gives

$$\tilde{H}_P = \sum_{m>n} \hat{h}_{mn}e^{-i\omega_{mn}t} + \hat{h}^\dagger_{mn}e^{i\omega_{mn}t}. \tag{S35}$$

With $\frac{1}{\bar{\omega}_{mn,jk}} = \frac{1}{2}\left(\frac{1}{\omega_{mn}} + \frac{1}{\omega_{jk}}\right)$, the effective Hamiltonian describing the pair production processes found using perturbation theory is:

$$\hat{H}_P = \sum_{m>n,j>k} \frac{1}{\bar{\omega}_{mn,jk}}[\hat{h}^\dagger_{mn}, \hat{h}_{jk}] \exp^{i\left(\omega_{mn}-\omega_{jk}\right)t}. \tag{S36}$$

The dominant contributions come from $[\hat{h}^\dagger_{1,0}, \hat{h}_{1,0}]$ and $[\hat{h}^\dagger_{0,-1}, \hat{h}_{0,-1}]$ (first two processes in Fig. S2B), and gives an approximate result

$$\hat{H}_P \approx \frac{\chi^2}{8\omega_{rec}}\hat{S}^+_0\hat{S}^-_0(\hat{S}^z_{-1} - \hat{S}^z_1) = \frac{\chi^2 N}{8\omega_{rec}}\hat{S}^+_0\hat{S}^-_0, \tag{S37}$$

with $\chi_{1,0}, \chi_{0,-1} \approx \chi_+ + \chi_-$. Thus the pair production processes at lowest order simply rescale the momentum-exchange interaction strength as $\chi \rightarrow \chi\left(1 + \frac{\chi N}{8\omega_{rec}}\right)$. For most of the experimental results presented in the paper, the collective interaction strength χN is less than $2\pi \times 1.5$ kHz such that the fractional correction is smaller than 5%. For the highest interaction strength we applied (red trace in Fig. 4A), the fractional correction is only 12%.

3.5 Mean-field dynamics for the full atom-cavity system

Here we give a different but more physical method to derive the system dynamics. Start from full atom-cavity Hamiltonian Eq. (S16), the equation of motions for the collective atomic operators is given by:

$$\left\langle \dot{\hat{J}}_+ \right\rangle = i\omega_z \left\langle \hat{J}_+ \right\rangle - i\frac{g_0^2}{2\Delta_a} \langle \hat{a}^\dagger \hat{a} \rangle \left\langle \hat{J}_z \right\rangle ,$$
$$\left\langle \dot{\hat{J}}_- \right\rangle = -i\omega_z \left\langle \hat{J}_- \right\rangle + i\frac{g_0^2}{2\Delta_a} \langle \hat{a}^\dagger \hat{a} \rangle \left\langle \hat{J}_z \right\rangle , \tag{S38}$$
$$\left\langle \dot{\hat{J}}_z \right\rangle = -i\frac{g_0^2}{4\Delta_a} \langle \hat{a}^\dagger \hat{a} \rangle \left(\left\langle \hat{J}_+ \right\rangle - \left\langle \hat{J}_- \right\rangle \right) .$$

We further define $\tilde{\hat{J}}_+ = \hat{J}_+ e^{-i\omega_z t}$ and $\tilde{\hat{J}}_- = \hat{J}_- e^{i\omega_z t}$, where $\tilde{\hat{J}}_\pm$ are slowly varying and thus assumed to be constants during the time scale set by Δ_d and ω_z. Then the mean-field equations of motion for the cavity field operators are

$$\frac{d}{dt} \langle \hat{a} \rangle = i(\Delta_d + i\kappa/2) \langle \hat{a} \rangle - i\alpha_d \sqrt{\kappa_1} - i\frac{g_0^2}{4\Delta_a} \left(\left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} + \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \right) \langle \hat{a} \rangle ,$$
$$\frac{d}{dt} \left\langle \hat{a}^\dagger \hat{a} \right\rangle = i \left(\alpha_d^* \sqrt{\kappa_1} \langle \hat{a} \rangle - \alpha_d \sqrt{\kappa_1} \left\langle \hat{a}^\dagger \right\rangle \right) - \kappa \left\langle \hat{a}^\dagger \hat{a} \right\rangle , \tag{S39}$$

where the atomic properties $\left\langle \tilde{\hat{J}}_\pm \right\rangle$ modulate the cavity field and thus create modulation side-bands that are crucial for the exchange interactions. The formal integration for the cavity field is given by

$$\langle \hat{a} \rangle_t = -i\alpha_d \sqrt{\kappa_1} \exp\left[i \int_0^t d\tau \, \mathcal{C}(\tau) \right] \int_0^t dt' \, \exp\left[-i \int_0^{t'} d\tau \, \mathcal{C}(\tau) \right], \tag{S40}$$

with

$$\mathcal{C}(t) = \Delta_d + i\frac{\kappa}{2} - \frac{g_0^2}{4\Delta_a} \left(\left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} + \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \right)$$
$$= \Delta_d + i\frac{\kappa}{2} - \frac{g_0^2}{4\Delta_a} \left(\left\langle \tilde{\hat{J}}_x \right\rangle \cos \omega_z t - \left\langle \tilde{\hat{J}}_y \right\rangle \sin \omega_z t \right) . \tag{S41}$$

We then have:

$$
\begin{aligned}
\exp\left[i\int_0^t d\tau \mathcal{C}(\tau)\right] =& e^{i(\Delta_d+i\kappa/2)t}\exp\left[i\frac{g_0^2\left\langle\tilde{\hat{J}}_y\right\rangle}{2\Delta_a\omega_z}\right]\exp\left[-i\frac{g_0^2}{2\Delta_a\omega_z}\left[\left\langle\tilde{\hat{J}}_x\right\rangle\sin(\omega_z t)+\left\langle\tilde{\hat{J}}_y\right\rangle\cos(\omega_z t)\right]\right]\\
=& e^{i(\Delta_d+i\kappa/2)t}\exp\left[i\frac{g_0^2\left\langle\tilde{\hat{J}}_y\right\rangle}{2\Delta_a\omega_z}\right]\sum_{n=-\infty}^{+\infty}(-1)^n J_n\left(\frac{g_0^2\left\langle\tilde{\hat{J}}_x\right\rangle}{2\Delta_0\omega_z}\right)e^{in\omega_z t}\\
&\times\sum_{m=-\infty}^{+\infty}(-i)^m J_m\left(\frac{g_0^2\left\langle\tilde{\hat{J}}_y\right\rangle}{\Delta_0\omega_z}\right)e^{im\omega_z t}\\
=& e^{i(\Delta_d+i\kappa/2)t}\exp\left[i\frac{g_0^2\left\langle\tilde{\hat{J}}_y\right\rangle}{2\Delta_a\omega_z}\right]\sum_{n=-\infty}^{+\infty}(-1)^n J_n\left(\frac{g_0^2\left|\left\langle\tilde{\hat{J}}_+\right\rangle\right|}{2\Delta_a\omega_z}\right)e^{in\theta}e^{in\omega_z t}\\
=& e^{i(\Delta_d+i\kappa/2)t}e^{i\Theta},
\end{aligned}
\tag{S42}
$$

where $\tan\theta=\left\langle\tilde{\hat{J}}_y\right\rangle/\left\langle\tilde{\hat{J}}_x\right\rangle$, and $J_n(z)$ is the Bessel function of the first kind. In the derivation above, we use the Bessel function properties:

$$
e^{iz\cos\phi}=\sum_{n=-\infty}^{+\infty}i^n J_n(z)e^{in\phi},\quad e^{iz\sin\phi}=\sum_{n=-\infty}^{+\infty}J_n(z)e^{in\phi},
\tag{S43}
$$

as well as

$$
e^{i\nu\phi}J_\nu(R)=\sum_{k=-\infty}^{\infty}J_k(\rho)J_{\nu+k}(r)e^{ik\varphi},\quad R=\sqrt{r^2+\rho^2-2r\rho\cos\varphi},\quad e^{2i\phi}=\frac{r-\rho e^{-i\varphi}}{r-\rho e^{i\varphi}}.
\tag{S44}
$$

Similarly, we have:

$$
\begin{aligned}
\int_0^t dt'\exp\left[-i\int_0^{t'}d\tau\mathcal{C}(\tau)\right] =& i\exp\left[-i\frac{g_c^2\left\langle\tilde{\hat{J}}_y\right\rangle}{\Delta_0\omega_z}\right]\sum_{n=-\infty}^{+\infty}(-1)^n J_n\left(\frac{g_c^2\left|\left\langle\tilde{\hat{J}}_+\right\rangle\right|}{\Delta_0\omega_z}\right)\\
&\times e^{-in\theta}\frac{e^{-i(\Delta_d+i\kappa/2+n\omega_z)t}-1}{\Delta_d+i\kappa/2+n\omega_z}.
\end{aligned}
\tag{S45}
$$

In the limit $g_c^2N/4\Delta_0\ll\omega_z$, we expand to the first order of the Bessel function and solve the cavity field at long time with $t \gg 1/\kappa$

$$\langle \hat{a} \rangle_t \approx \frac{\alpha_d \sqrt{\kappa_1}}{\Delta_d + i\kappa/2} e^{i\Theta} \left[1 + \frac{g_0^2 \left\langle \tilde{\hat{J}}^{+} \right\rangle}{4\Delta_a \omega_z} \frac{\Delta_d + i\kappa/2}{\Delta_d - \omega_z + i\kappa/2} e^{i\omega_z t} - \frac{g_0^2 \left\langle \tilde{\hat{J}}^{-} \right\rangle}{4\Delta_a \omega_z} \frac{\Delta_d + i\kappa/2}{\Delta_d + \omega_z + i\kappa/2} e^{-i\omega_z t} \right]. \tag{S46}$$

From Eq. (S39), the formal integration for $\left\langle \hat{a}^\dagger \hat{a} \right\rangle$ is given by,

$$\left\langle \hat{a}^\dagger \hat{a} \right\rangle_t = e^{-\kappa t} \int_0^t d\tau \; i e^{\kappa \tau} \sqrt{\kappa_1} \left(\alpha_d^* \langle \hat{a} \rangle_\tau - \alpha_d \left\langle \hat{a}^\dagger \right\rangle_\tau \right). \tag{S47}$$

With Eq. (S46), we have:

$$\begin{aligned}
\left\langle \hat{a}^\dagger \hat{a} \right\rangle_t = |\alpha_0|^2 &+ \frac{g_0^2 |\alpha_d|^2 \kappa_1}{4\Delta_a \omega_z} \left[\frac{1}{\left(\Delta_d - \omega_z + \frac{i\kappa}{2}\right)\left(\Delta_d - \frac{i\kappa}{2}\right)} - \frac{1}{\left(\Delta_d + \omega_z - \frac{i\kappa}{2}\right)\left(\Delta_d + \frac{i\kappa}{2}\right)} \right] \left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} \\
&+ \frac{g_0^2 |\alpha_d|^2 \kappa_1}{4\Delta_a \omega_z} \left[\frac{1}{\left(\Delta_d - \omega_z - \frac{i\kappa}{2}\right)\left(\Delta_d + \frac{i\kappa}{2}\right)} - \frac{1}{\left(\Delta_d + \omega_z + \frac{i\kappa}{2}\right)\left(\Delta_d - \frac{i\kappa}{2}\right)} \right] \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \\
&\equiv |\alpha_0|^2 + A \left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} + A^* \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t}.
\end{aligned} \tag{S48}$$

Substituting the solution $\left\langle \hat{a}^\dagger \hat{a} \right\rangle_t$ in Eq. (S38), here are the equations of motion with only atom-atom interaction

$$\begin{aligned}
\left\langle \dot{\hat{J}}_+ \right\rangle &= i\omega_z \left\langle \hat{J}_+ \right\rangle - i\frac{g_0^2}{2\Delta_a} \left(|\alpha_0|^2 + A \left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} + A^* \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \right) \left\langle \hat{J}_z \right\rangle, \\
\left\langle \dot{\hat{J}}_- \right\rangle &= -i\omega_z \left\langle \hat{J}_- \right\rangle + i\frac{g_0^2}{2\Delta_a} \left(|\alpha_0|^2 + A \left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} + A^* \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \right) \left\langle \hat{J}_z \right\rangle, \\
\left\langle \dot{\hat{J}}_z \right\rangle &= -i\frac{g_0^2}{4\Delta_a} \left(|\alpha_0|^2 + A \left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} + A^* \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \right) \left(\left\langle \tilde{\hat{J}}_+ \right\rangle e^{i\omega_z t} - \left\langle \tilde{\hat{J}}_- \right\rangle e^{-i\omega_z t} \right).
\end{aligned} \tag{S49}$$

With the slowly varying variables $\left\langle \tilde{\hat{J}}_\pm \right\rangle$, the above equation of motion can be further simplified:

$$\begin{aligned}
\left\langle \dot{\tilde{\hat{J}}}_+ \right\rangle &= -i\frac{g_0^2}{2\Delta_a} A \left\langle \tilde{\hat{J}}_+ \right\rangle \left\langle \hat{J}_z \right\rangle, \\
\left\langle \dot{\tilde{\hat{J}}}_- \right\rangle &= i\frac{g_0^2}{2\Delta_a} A^* \left\langle \tilde{\hat{J}}_- \right\rangle \left\langle \hat{J}_z \right\rangle, \\
\left\langle \dot{\hat{J}}_z \right\rangle &= -\frac{g_0^2}{2\Delta_a} \frac{A - A^*}{2i} \left\langle \tilde{\hat{J}}_+ \right\rangle \left\langle \tilde{\hat{J}}_- \right\rangle,
\end{aligned} \tag{S50}$$

where the single-particle processes (i.e. off-resonant Bragg coupling) and the fast oscillating dynamics at frequency $2\omega_z$ are ignored. With $g_0^2 A/2\Delta_0 = 2(\chi_+ + \chi_-) + i(\Gamma_+ - \Gamma_-)$, the above equations of motion agree with Eq. (S29) and thus verify the predicted dynamics from the effective spin Hamiltonian. In the limit without superradiance, $\left\langle \dot{\hat{J}}_z \right\rangle = 0$ such that the total spin projection is conserved.

3.6 Exchange interaction versus one-axis twisting

In the absence of inhomogeneity, the exchange interaction predicts the same dynamics for the collective operators as the one-axis twisting interaction (1,12). To emphasize the contrast between the two, we now consider a minimal model for two spin ensembles $\hat{J}_1$ and $\hat{J}_2$ with a differential transition frequency δ, in light of Fig. 4(C) with two spin ensembles defined for $p > 0$ and $p < 0$.

For the exchange interaction with the Hamiltonian $\hat{H} = \hbar\chi \left(\hat{J}_{1+} + \hat{J}_{2+}\right)\left(\hat{J}_{1-} + \hat{J}_{2-}\right) + \frac{\hbar\delta}{2}\left(\hat{J}_{1z} - \hat{J}_{2z}\right)$, here are the equations of motion for individual spin operators

$$
\begin{aligned}
\left\langle \dot{\hat{J}}_{1+} \right\rangle &= -i2\chi \left(\left\langle \hat{J}_{1+} \right\rangle + \left\langle \hat{J}_{2+} \right\rangle \right) \left\langle \hat{J}_{1z} \right\rangle + i\delta \left\langle \hat{J}_{1+} \right\rangle, \\
\left\langle \dot{\hat{J}}_{2+} \right\rangle &= -i2\chi \left(\left\langle \hat{J}_{1+} \right\rangle + \left\langle \hat{J}_{2+} \right\rangle \right) \left\langle \hat{J}_{2z} \right\rangle - i\delta \left\langle \hat{J}_{2+} \right\rangle, \\
\left\langle \dot{\hat{J}}_{1z} \right\rangle &= i\chi \left(\left\langle \hat{J}_{1-} \right\rangle \left\langle \hat{J}_{2+} \right\rangle - \left\langle \hat{J}_{1+} \right\rangle \left\langle \hat{J}_{2-} \right\rangle \right), \\
\left\langle \dot{\hat{J}}_{2z} \right\rangle &= i\chi \left(\left\langle \hat{J}_{1+} \right\rangle \left\langle \hat{J}_{2-} \right\rangle - \left\langle \hat{J}_{1-} \right\rangle \left\langle \hat{J}_{2+} \right\rangle \right),
\end{aligned}
\tag{S51}
$$

from which we can clearly see that the inhomogenity term causes oscillations for the coherence $\left\langle \hat{J}_{1+} \right\rangle$ and $\left\langle \hat{J}_{2+} \right\rangle$ and population differences $\left\langle \hat{J}_{1z} \right\rangle$ and $\left\langle \hat{J}_{2z} \right\rangle$ as discussed in Fig. 4(C). Also, with the exchange interaction, the total spin projection $\left\langle \hat{J}_z \right\rangle$ converses given $\frac{d\langle \hat{J}_z \rangle}{dt} = \frac{d\langle \hat{J}_{1z} \rangle}{dt} + \frac{d\langle \hat{J}_{2z} \rangle}{dt} = 0$, while the individual spin projections $\left\langle \hat{J}_{1z} \right\rangle$ and $\left\langle \hat{J}_{2z} \right\rangle$ are not conserved.

In contrast, the equations of motion for OAT with inhomogenity $\hat{H} = \hbar\chi \left(\hat{J}_{1z} + \hat{J}_{2z}\right)^2 +$

$\frac{\hbar\delta}{2}\left(\hat{J}_{1z}-\hat{J}_{2z}\right)$ are

$$\begin{aligned}\left\langle\dot{\hat{J}}_{1+}\right\rangle &= -i2\chi\left(\left\langle\hat{J}_{1+}\right\rangle+\left\langle\hat{J}_{2+}\right\rangle\right)\left\langle\hat{J}_{1z}\right\rangle,\\ \left\langle\dot{\hat{J}}_{2+}\right\rangle &= -i2\chi\left(\left\langle\hat{J}_{1+}\right\rangle+\left\langle\hat{J}_{2+}\right\rangle\right)\left\langle\hat{J}_{2z}\right\rangle.\end{aligned} \tag{S52}$$

Because the inhomogenity term commutes with the interaction term, both the total and individual spin projections $\left\langle\hat{J}_z\right\rangle$, $\left\langle\hat{J}_{1z}\right\rangle$ and $\left\langle\hat{J}_{2z}\right\rangle$ are conserved.

For all the simulation results shown in the main text, we account for the effect of Doppler broadening (inhomogeneous kinetic energy difference) by including $\hat{H}_{\text{in}}$ for all the momentum states in the equations of motion and solving the equations of motion numerically.

4 Collective recoil and Suppression of Doppler Dephasing

In the main text, we point out the unusualness of the momentum-exchange interaction with a *gedanken* experiment. Here, we expand and analyze this observation from three different perspectives. We first focus on the case invoked in the main text with only one atom undergoing the two-photon transition and elucidate the collective recoil with an analytical derivation. We then generalize it to allow for more initial recoiling atoms using mean-field numerical simulations of the wave packet dynamics. The simulations explicitly show the the collective recoil of the total atomic wave packet which results in the enhanced coherence observed in the Ramsey-type experiment in the main text.

In Mössbauer spectroscopy, Doppler broadening of a Rabi spectrum is suppressed because the whole crystal collective recoils rather than just a single photon-absorbing atom embedded in the crystal (34). We present numerical simulations that show that Doppler broadening of a Rabi spectrum is also suppressed by the gap protection term $\hat{J}^2$.

4.1 *Gedanken* experiment: absorbing exactly one photon

We first consider initializing the system with N atoms having the same momentum distributions centered at $p_0 - \hbar k$. The quantum state of the system can be expressed in second quantization as $|\psi_{ini}\rangle = \frac{1}{\sqrt{N!}}\left(\int_{-\hbar k}^{+\hbar k}\phi(p)\hat{\psi}^\dagger(p)dp\right)^N|vac\rangle$ with $|vac\rangle$ the vacuum state in which there are no atom present and $\phi(p) \propto \exp\{-p^2/4\sigma_p^2\}$ the probability amplitude distribution with rms spread in probability σ_p. We then introduce a different basis set $|N_a, N_b\rangle$ in which N_a denote the number of atoms with momentum $\in (p_0 - 2\hbar k, p_0)$ and N_b atoms the number of atoms with momentum $\in (p_0, p_0 + 2\hbar k)$. In this new basis, the initial state can be expressed as $|\psi_{ini}\rangle = |N, 0\rangle$.

The *gedanken* experiment described in the main text considers a single photon combined with a coherent state of light driving the two-photon Bragg transition. One can not tell which atom absorbed the single photon and underwent the two-photon transition, such that the state should be symmetrized with respect to which atom experiences the $2\hbar k$ momentum transfer. The resulting state can be expressed as

$$|\psi_0\rangle = |N-1, 1\rangle = \frac{\sum_p \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p)}{\sqrt{N}}|N, 0\rangle\,, \tag{S53}$$

with the sum implicitly over discrete states between $\pm\hbar k$. This initial state then evolves under the Hamiltonian $\hat{H} = \hat{H}_{kin} + \hat{H}_{int}$:

$$\hat{H}_{kin} = \sum_p \frac{(p_0 - \hbar k + p)^2}{2m}\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p) + \sum_p \frac{(p_0 + \hbar k + p)^2}{2m}\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p) \tag{S54}$$

$$\hat{H}_{int} = \chi\sum_{p,q}\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p)\hat{\psi}_\downarrow^\dagger(q)\hat{\psi}_\uparrow(q) \tag{S55}$$

The first term is the kinetic energy and second the exchange-interaction.

Without applying the interaction ($\chi = 0$), after time evolution the final state becomes

$$|\psi_{free}(t)\rangle = \hat{U}_0|\psi_0\rangle\,, \tag{S56}$$

with $\hat{U}_0 = \exp\left[-i\hat{H}_{kin}t/\hbar\right]$. We can then calculate the spatial probability distribution:

$$\langle\psi_{free}(t)|\,\hat{c}_Z^\dagger\hat{c}_Z\,|\psi_{free}(t)\rangle \propto (N-1)\exp\left[-\frac{\left(Z-\frac{p_0-\hbar k}{m}t\right)^2}{2\sigma_Z^2(t)}\right]+\exp\left[-\frac{\left(Z-\frac{p_0+\hbar k}{m}t\right)^2}{2\sigma_Z^2(t)}\right], \tag{S57}$$

where the $\hat{c}_Z^\dagger$ is the operator for creating an atom at position Z defined by $\hat{c}_Z^\dagger \propto \sum_p \hat{\psi}_\downarrow^\dagger(p)e^{-i(p_0-\hbar k+p)Z/\hbar} + \sum_p \hat{\psi}_\uparrow^\dagger(p)e^{-i(p_0+\hbar k+p)Z/\hbar}$, and $\sigma_Z(t) = \frac{\hbar}{2\sigma_p}\sqrt{1+\left(\frac{2\sigma_p^2 t}{m\hbar}\right)^2}$ is the rms spread of the wave packets in position space. The result above corresponds to two Gaussian distributions moving at different velocities with heights proportional to the number of atoms in each momentum distribution centered at $p_0 \pm \hbar k$.

Now we turn to the case with interaction applied. The initial Dicke-like state $|\psi_0\rangle$ of Eq. S53 is an eigenstate of $\hat{H}_{int}$ with eigenvalue χN as discussed in Sec. 4.4. On the other hand, $\hat{H}_{kin}$ couples the symmetrized Dicke-like state to other eigenstates of the interaction Hamiltonian that are separated by an energy gap χN (3,27). In the limit $\chi N \gg \sigma_{in}$, we can consider $\hat{H}_{kin}$ as a perturbation of $\hat{H}_{int}$, with the time evolution being governed by the projection of $\hat{H}_{kin}$ onto the initial Dicke-like state with operator $\hat{P} = |N-1,1\rangle\langle N-1,1|$ (see details of the derivation in Sec. 4.4)

$$|\psi_{int}(t)\rangle = \hat{U}_0'\,|\psi_0\rangle\,, \tag{S58}$$

with $\hat{U}_0' = \exp\left[-i\hat{P}\hat{H}_{kin}\hat{P}t/\hbar\right]$. We can then calculate the spatial probability distribution:

$$\langle\psi_{int}(t)|\,\hat{c}_Z^\dagger\hat{c}_Z\,|\psi_{int}(t)\rangle \propto \exp\left[-\frac{\left(Z-\frac{\bar{p}t}{m}\right)^2}{2\sigma_Z^2(t)}\right], \tag{S59}$$

which corresponds to a single Gaussian wave packet moving at the velocity $\bar{p}/m = \frac{N-1}{N}\frac{p_0-\hbar k}{m} + \frac{1}{N}\frac{p_0+\hbar k}{m}$. Relative to the co-moving frame of the atoms before absorbing the single photon, this corresponds to a velocity of $\frac{1}{N}\frac{p_0+\hbar k}{m}$. Thus the momentum-exchange interaction leads to collective recoil of the atomic ensemble at a velocity reduced by $1/N$.

4.2 Generalized *gedanken* experiment: absorbing $\bar{M}$ photons

We now consider the case where N atoms starts with a momentum distribution centered at $p_0 - \hbar k$ and a coherent two-photon Bragg coupling is applied to place the atoms in a superposition of momentum states such that on average $\bar{M}$ atoms undergo the two-photon transition. In this scenario, the Bragg pulse transfers $\bar{M}2\hbar k$ of total momentum to the atomic ensemble. In Fig. S3, we perform mean-field numerical simulations of the wave-packet spatial probability distribution without interactions and with a Heisenberg interaction $\hat{J}^2$ provided by the momentum-exchange interactions but neglecting the Ising or OAT contribution $\hat{J}_z^2$. We consider three initial Bragg pulses areas ($\pi/4, \pi/2$ and $3\pi/4$) to achieve $\bar{M}/N \approx 0.15, 0.5$ and 0.85 and simulate the evolution of the spatial probability distributions of the atoms as a function of time of flight period T_{TOF} after the initial Bragg pulse.

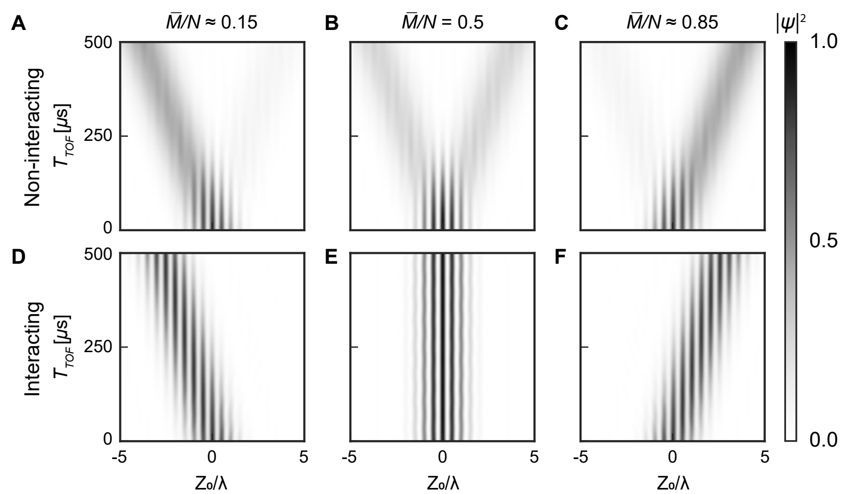

Figure S3: **Simulated evolution of the wave packets.** Evolution of the spatial probability distributions in the moving frame defined by the p_0 state with $Z_0 = Z - \frac{p_0}{m}t$. With all atoms initialized in $p_0 - \hbar k$ momentum state, $\pi/4$, $\pi/2$ and $3\pi/4$ Bragg pulses are applied to achieve different fractions of the initial recoiling atoms. With no interaction applied (χ=0), the interference fringes start to disappear as the wave packets separate into two for $\bar{M}/N$ =0.25 (A), 0.5 (B) and 0.75 (C). As a comparison, with the interaction applied ($\chi_{\pm}N \gg \sigma_{in}$), the wave packets do not separate but instead move at an averaged velocity determined by $\bar{M}/N$ as shown in (D), (E) and (F).

As shown in Fig. S3 (A) to (C), without the interaction ($\chi N = 0$), the wave packets separate into two with increasing T_{TOF}, which also leads to the vanishing interference fringes. The disappearance of the interference fringes caused by the two wave packets separating is precisely Doppler dephasing in this wave-packet picture. With the interaction applied ($\chi N \gg \sigma_{in}$), as shown in Fig. S3 (D) to (F), the wave packets no longer separate but instead move at velocity proportional to $\bar{M}/N$. Note that the interference fringes now persist because the atoms collectively recoil as a single wave packet. This is why the collective recoil mechanism leads to a suppression of the Doppler dephasing.

One also notices that the fringes are stationary in the reference frame of Fig S3, independent of $\bar{M}/N$. This implies that the fringes always move at a velocity p_0/m in the lab frame. If one considers an acceleration a_Z (such as due to gravity), the momentum p_0 now goes to $p_0 \rightarrow p_0 + m\, a_Z\, t$. The changing velocity of the interference fringes provides the interferometric sensitivity to accelerations, even in the presence of the Heisenberg interaction $\hat{J}^2$.

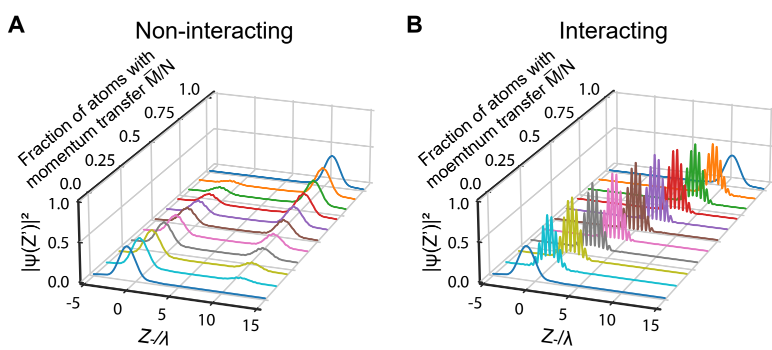

Figure S4: **Simulation of collective recoil.** Simulated spatial probability distributions of the wave functions after evolution in the moving frame defined by the $p_0 - \hbar k$ state with $Z_- = Z - \frac{p_0 - \hbar k}{m} t$. **(A)** Without the interaction applied, the wave packets split into two regardless of the fraction of initial recoiled atoms. **(B)**After turning on the interaction, the wave packets no longer separate but instead recoil together with an averaged momentum across the whole ensemble.

By fixing the time of flight period $T_{TOF} = 500\ \mu$s, we can also compare the results with different fractions of initial recoiling atoms $\bar{M}/N$. As shown in Fig. S4, when there is no interaction applied, as one expects, the atoms are only found in two spatially distinct regions corresponding to either having not absorbed two-photon recoil or having absorbed two-photon recoil and gaining a velocity $v_{rec} = 2\hbar k/m$. Varying the number of two-photon momentum kicks $\bar{M}$ only varies the fraction of atoms found in one spatial region or the other, but not the centers of the two distinct spatial regions.

However, when the momentum-exchange interaction is much larger than the Doppler broadening $|\chi N| \gg \sigma_{in}$, one sees in Fig. S4B that now the atoms recoil as a single wave packet with wave packet velocity $v = \frac{\bar{M}}{N} v_{rec}$ that depends on the fraction of atoms $\bar{M}/N$ that underwent two-photon absorption. For simplicity, the simulations above included only the gap protection term $\hat{J}^2$ and neglected the Ising term $\hat{J}_z^2$ as this latter term only shifts the phase of the interference fringes within the wavepacket, but does not alter the collective wavepacket recoil velocity.

4.3 Simulation of Rabi spectroscopy line narrowing

In the main text, it was experimentally shown that the coherence of the system is extended in time for a Ramsey sequence. The Rabi-spectroscopy line is also narrowed below the Doppler broadening linewidth by the gap protection term $\hat{J}^2$. In Fig. S5, we show the results of simulated Rabi-spectroscopy in which the Rabi frequency of the Bragg coupling is chosen to be much smaller than the Doppler broadening, and the Bragg coupling is applied to form a π-pulse for atoms resonant with the coupling. The black curve shows the probability of undergoing a two-photon transition to another momentum state with the width reflecting the Doppler broadening of the ensemble. The red curve shows the same Rabi spectroscopy scan but with $\chi N \gg \sigma_{in}$. The feature is both narrower, primarily reflecting the time for which the coupling is applied, and is higher since all atoms undergo their transition at the same frequency. We cannot observe this direct narrowing in the experiment due to the additional OAT or Ising interaction $\hat{J}_z^2$ produced by the momentum exchange, since the $\hat{J}_z$ changes as a function of time during the

Rabi probing. However we do observe the narrowing via the extension of coherence time in a Ramsey measurement since the spin projection J_z is fixed during the Ramsey evolution time. In principle, the OAT or Ising interaction could be canceled by toggling the sign of χ via changing the dressing laser frequency during the Ramsey evolution period, while the gap protection is preserved since it does not depend on the sign of χ.

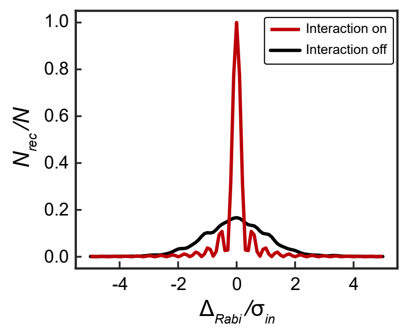

Figure S5: **Simulation of Rabi spectroscopy.** Fraction of the atoms undergoing the transition from $p_0 - \hbar k$ to $p_0 + \hbar k$ as a function of the detuning Δ_{Rabi} of the Bragg pulse from the transition frequency ω_z. The linewidth of the spectroscopic signal reflects the Doppler broadening when interactions are not applied (black trace). With the interaction applied, the spectroscopic linewidth is now narrowed (red trace) due to the suppression of Doppler dephasing by the collective recoil.

4.4 Detailed derivation of the analytic results

A detailed derivation of the analytic results in Sec. 4.1 is presented here. We first define the operator for creating a single particle with momentum $\in (p_0 - 2\hbar k, p_0)$:

$$\hat{A}^\dagger = \sum_p \phi(p)\hat{\psi}_\downarrow^\dagger(p), \tag{S60}$$

where $\phi(p) = \sqrt{\mathcal{N}} e^{-p^2/4\sigma_p^2}$ with $\mathcal{N} = 1/\sum_p e^{-p^2/2\sigma_p^2}$ to satisfy the commutation relation:

$$\left[\hat{A}, \hat{A}^\dagger\right] = 1, \tag{S61}$$

which gives

$$\left[\hat{\psi}_{\downarrow}(p), \hat{A}^{\dagger}\right] = \phi(p), \quad \left[\hat{\psi}_{\downarrow}(p), \left(\hat{A}^{\dagger}\right)^{N}\right] = N\phi(p)\left(\hat{A}^{\dagger}\right)^{N-1}, \quad \left[\hat{\psi}_{\uparrow}(q), \hat{A}^{\dagger}\right] = 0. \tag{S62}$$

The aforementioned Fock state, describing N atoms all in the lower momentum window (centered at $p_0 - \hbar k$) is given by:

$$|N, 0\rangle = \frac{\left(\hat{A}^{\dagger}\right)^{N}}{\sqrt{N!}} |vac\rangle = \sum_{\vec{n}} A_{\vec{n}} |\vec{n}\rangle_{G}. \tag{S63}$$

Here, $|\vec{n}\rangle_G = |n_{p_1}, n_{p_2}, \cdots\rangle$ represents n_{p_i} atoms (with $\sum_{p_i} n_{p_i} = N$) occupying discrete momentum mode p_i in the lower momentum window centered at $p_0 - \hbar k$ and no atom in the higher momentum window. Applying the operator $\hat{\psi}_{\downarrow}(p)$ to $|N, 0\rangle$, we find:

$$\begin{aligned}
\hat{\psi}_{\downarrow}(p) |N, 0\rangle &= \frac{\hat{\psi}_{\downarrow}(p)\left(\hat{A}^{\dagger}\right)^{N}}{\sqrt{N!}} |vac\rangle \\
&= \frac{\left(\hat{A}^{\dagger}\right)^{N}\hat{\psi}_{\downarrow}(p) + N\phi(p)\left(\hat{A}^{\dagger}\right)^{N-1}}{\sqrt{N!}} |vac\rangle \\
&= \phi(p)\sqrt{N} |N-1, 0\rangle .
\end{aligned} \tag{S64}$$

The Dicke-like state generated after the absorption of a photon can be obtained by applying the operator $\sum_p \hat{\psi}_{\uparrow}^{\dagger}(p)\hat{\psi}_{\downarrow}(p)$ to $|N, 0\rangle$. This creates a single excitation :

$$|N-1, 1\rangle = \frac{\sum_p \hat{\psi}_{\uparrow}^{\dagger}(p)\hat{\psi}_{\downarrow}(p)}{\sqrt{N}} |N, 0\rangle = \sum_{\vec{n}} A_{\vec{n}} \left|\vec{n'}\right\rangle_{D}. \tag{S65}$$

where $\left|\vec{n'}\right\rangle_{D} = \frac{\sum_p \hat{\psi}_{\uparrow}^{\dagger}(p)\hat{\psi}_{\downarrow}(p)}{\sqrt{N}} |\vec{n}\rangle_G$ and the normalization factor $1/\sqrt{N}$ comes from the fact that:

$${}_G\langle\vec{n}| \sum_q \hat{\psi}_{\downarrow}^{\dagger}(q)\hat{\psi}_{\uparrow}(q) \sum_p \hat{\psi}_{\uparrow}^{\dagger}(p)\hat{\psi}_{\downarrow}(p) |\vec{n}\rangle_G = \sum_p {}_G\langle\vec{n}|\hat{\psi}_{\downarrow}^{\dagger}(p)\hat{\psi}_{\downarrow}(p) |\vec{n}\rangle_G = N. \tag{S66}$$

The observable we want to compute is the time dependent expectation value of $\hat{c}_Z^{\dagger}\hat{c}_Z$. Given that the real space creation operator is given by $\hat{c}_Z^{\dagger} \propto \sum_p \hat{\psi}_{\downarrow}^{\dagger}(p)e^{-i(p_0-\hbar k+p)Z/\hbar} + \sum_p \hat{\psi}_{\uparrow}^{\dagger}(p)e^{-i(p_0+\hbar k+p)Z/\hbar}$, the desired observable is given by

$$\hat{c}_Z^\dagger \hat{c}_Z \propto \sum_{pq} \left(\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q) + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q) + \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(q)e^{i2kZ} + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q)e^{-i2kZ} \right) e^{i(q-p)Z/\hbar}. \tag{S67}$$

We proceed to evaluate its expectation value for two different situations.

4.4.1 Evolution with no interaction

Without the interaction applied, the time evolution of the Dicke-like state is solely described by the unitary operator $\hat{U}_0$, which gives $|\psi_{free}(t)\rangle = \hat{U}_0 |\psi_0\rangle$. We can compute from it separately the different spatial distribution profiles:$\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q)$, $\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q)$, $\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(q)$ and $\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q)$.

By using the property $\hat{\psi}_\uparrow(l)\hat{\psi}_\uparrow^\dagger(k)|N,0\rangle = \delta_{k,l}|N,0\rangle$ and Eq. (S64),

$$\begin{aligned}
&\langle\psi_{free}(t)|\, \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q)\, |\psi_{free}(t)\rangle \\
=&\frac{1}{N}\langle N,0| \left(\sum_l \hat{\psi}_\uparrow^\dagger(l)\hat{\psi}_\downarrow(l)\right)^\dagger \hat{U}_0^\dagger \left(\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q)\right) \hat{U}_0 \left(\sum_s \hat{\psi}_\uparrow^\dagger(s)\hat{\psi}_\downarrow(s)\right) |N,0\rangle \\
=&(N-1)\, e^{i\frac{(p+p_0-\hbar k)^2-(q+p_0-\hbar k)^2}{2m\hbar}t}\phi(p)\phi(q).
\end{aligned} \tag{S68}$$

Similarly,

$$\begin{aligned}
&\langle\psi_{free}(t)|\, \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q)\, |\psi_{free}(t)\rangle \\
=&\frac{1}{N}\langle N,0| \left(\sum_l \hat{\psi}_\uparrow^\dagger(l)\hat{\psi}_\downarrow(l)\right)^\dagger \hat{U}_0^\dagger \left(\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q)\right) \hat{U}_0 \left(\sum_s \hat{\psi}_\uparrow^\dagger(s)\hat{\psi}_\downarrow(s)\right) |N,0\rangle \\
=&e^{i\frac{(p+p_0+\hbar k)^2-(q+p_0+\hbar k)^2}{2m\hbar}t}\phi(p)\phi(q).
\end{aligned} \tag{S69}$$

Finally, we find $\langle\psi_{free}(t)|\, \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q)\, |\psi_{free}(t)\rangle = \langle\psi_{free}(t)|\, \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(q)\, |\psi_{free}(t)\rangle = 0$ due to the fact that the state $\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q)\, |\psi_{free}(t)\rangle$ and $|\psi_{free}(t)\rangle$ have different numbers of excitations in the higher momentum window and therefore do not overlap.

Summing all the terms together we obtain the spatial probability distribution :

$$
\begin{aligned}
&\langle\psi_{free}(t)|\, \hat{c}_Z^\dagger \hat{c}_Z \,|\psi_{free}(t)\rangle \\
\propto &\langle\psi_{free}(t)| \sum_{pq} \left(\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q) + \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q)\right) e^{i(q-p)Z/\hbar} \,|\psi_{free}(t)\rangle \\
=&\sum_{pq} \phi(p)\phi(q) e^{i(q-p)Z/\hbar} \left[(N-1)\, e^{i\frac{(p+p_0-\hbar k)^2-(q+p_0-\hbar k)^2}{2m\hbar}t} + e^{i\frac{(p+p_0+\hbar k)^2-(q+p_0+\hbar k)^2}{2m\hbar}t}\right] \\
=&\sum_{pq} \phi(p)\phi(q) e^{i(q-p)Z/\hbar} \left[(N-1)\, e^{i(p-q)\left(\frac{p+q}{2m}+\frac{p_0-k}{m}\right)t/\hbar} + e^{i(p-q)\left(\frac{p+q}{2m}+\frac{p_0+k}{m}\right)t/\hbar}\right] \\
\propto &(N-1)\exp\left\{\frac{-2\sigma_p^2\left(Z-\frac{p_0-\hbar k}{m}t\right)^2/\hbar^2}{1+16\left(\frac{t\sigma_p^2}{2m\hbar}\right)^2}\right\} + \exp\left\{\frac{-2\sigma_p^2\left(Z-\frac{p_0+\hbar k}{m}t\right)^2/\hbar^2}{1+16\left(\frac{t\sigma_p^2}{2m\hbar}\right)^2}\right\} \\
=&(N-1)\exp\left[-\frac{\left(Z-\frac{p_0-\hbar k}{m}t\right)^2}{2\sigma_Z^2(t)}\right] + \exp\left[-\frac{\left(Z-\frac{p_0+\hbar k}{m}t\right)^2}{2\sigma_Z^2(t)}\right],
\end{aligned}
\tag{S70}
$$

which corresponds to two different Gaussian wave packets moving at different velocities with on average $N-1$ atoms in one wave packet and only one atom in the other. Here, $\sigma_Z(t) = \frac{\hbar}{2\sigma_p}\sqrt{1+\left(\frac{2\sigma_p^2 t}{m\hbar}\right)^2}$ is the rms spread of the wave packets in position space, which increases over time due to the quadratic dispersion relationship.

4.4.2 Evolution with interactions

We now consider the time evolution of the Dicke-like state when the interaction is applied.

We first prove $\left|\vec{n'}\right\rangle_D$ is an eigenstate for $\hat{H}_{int}$ with eigenvalue χN:

$$
\begin{aligned}
\hat{H}_{int}\left|\vec{n'}\right\rangle_D &= \chi \sum_p \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p) \sum_q \hat{\psi}_\downarrow^\dagger(q)\hat{\psi}_\uparrow(q) \sum_k \hat{\psi}_\uparrow^\dagger(k)\hat{\psi}_\downarrow(k)\,|\vec{n}\rangle_G \\
&= \chi \sum_p \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p) \sum_{q,k} \hat{\psi}_\downarrow^\dagger(q)\hat{\psi}_\downarrow(k)\delta_{q,k}\,|\vec{n}\rangle_G \\
&= \chi \sum_p \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p) \sum_q \hat{\psi}_\downarrow^\dagger(q)\hat{\psi}_\downarrow(q)\,|\vec{n}\rangle_G \\
&= \chi N \sum_p \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(p)\,|\vec{n}\rangle_G \\
&= \chi N \left|\vec{n'}\right\rangle_D .
\end{aligned}
\tag{S71}
$$

On the other hand, $\hat{H}_{kin}$ couples the symmetrized Dicke-like state to other eigenstates of the interaction Hamiltonian that are separated by an energy gap χN (3,27). In the limit $\chi N \gg \sigma_{in}$, this coupling is energetically suppressed because of the large energy cost and to leading order we only need to account for the energy shift induced by $\hat{H}_{kin}$ when projected on $\left|\vec{n'}\right\rangle_D$:

$$
\begin{aligned}
&{}_D\left\langle \vec{n'} \middle| H_{kin} \middle| \vec{n'} \right\rangle_D \\
=&\frac{1}{N}\sum_p {}_G\langle\vec{n}| \sum_q \hat{\psi}_\downarrow(q)^\dagger \hat{\psi}_\uparrow(q) \left(\frac{(p+p_0-\hbar k)^2}{2m} \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p) + \frac{(p+p_0+\hbar k)^2}{2m} \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(p) \right) \\
&\quad \sum_k \hat{\psi}_\uparrow(k)^\dagger \hat{\psi}_\downarrow(k) \left|\vec{n}\right\rangle_G \\
=&\frac{1}{N}\sum_{pq} \frac{(p+p_0-\hbar k)^2}{2m} {}_G\left\langle\vec{n}\right| \hat{\psi}_\downarrow^\dagger(q)\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p)\hat{\psi}_\downarrow(q) \left|\vec{n}\right\rangle_G \\
&+ \frac{1}{N}\sum_p \frac{(p+p_0+\hbar k)^2}{2m} {}_G\left\langle\vec{n}\right| \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p) \left|\vec{n}\right\rangle_G
\end{aligned}
\tag{S72}
$$

Using the property ${}_G\langle\vec{n}|\hat{\psi}_\downarrow(q)^\dagger\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p)\hat{\psi}_\downarrow(q)\left|\vec{n}\right\rangle_G = n_p n_q\left(1-\delta_{p,q}\right) + n_p\left(n_p-1\right)\delta_{p,q}$, we find:

$$
\begin{aligned}
&{}_D\left\langle \vec{n'} \middle| H_{kin} \middle| \vec{n'} \right\rangle_D \\
=&\sum_{pq} \frac{(p+p_0-\hbar k)^2}{2m} \left(\frac{n_p n_q}{N}\left(1-\delta_{p,q}\right) + \delta_{p,q}\frac{n_p\left(n_p-1\right)}{N} \right) + \sum_p \frac{n_p}{N}\frac{(p+p_0+\hbar k)^2}{2m} \\
=&\sum_{pq} \frac{(p+p_0-\hbar k)^2}{2m} n_p \frac{n_q}{N} + \sum_p n_p \frac{1}{N}\left[\frac{(p+p_0+\hbar k)^2}{2m} - \frac{(p+p_0-\hbar k)^2}{2m} \right] \\
=&\sum_p n_p \left[\frac{(p+p_0-\hbar k)^2}{2m}\left(1-\frac{1}{N}\right) + \frac{(p+p_0+\hbar k)^2}{2m}\frac{1}{N} \right] \\
\equiv&\hbar\omega_{\vec{n}},
\end{aligned}
\tag{S73}
$$

noticing $\sum_q n_q/N = 1$ in the third line. Here, $\omega_{\vec{n}}$ is different for different momentum distribution described by $\vec{n}$.

The time-evolving state is then given by:

$$
\begin{aligned}
|\psi_{int}(t)\rangle &= \sum_{\vec{n}} e^{-i\omega_{\vec{n}}t} A_{\vec{n}} \left|\vec{n'}\right\rangle_D \\
&= \sum_{\vec{n}} A_{\vec{n}} e^{-i\sum_p n_p\left[\frac{(p+p_0-\hbar k)^2}{2m}\left(1-\frac{1}{N}\right)+\frac{(p+p_0+\hbar k)^2}{2m}\frac{1}{N}\right]t/\hbar} \left|\vec{n'}\right\rangle_D \\
&= \frac{1}{\sqrt{N}} \sum_q \hat{\psi}_\uparrow^\dagger(q)\hat{\psi}_\downarrow(q) e^{-i\sum_p \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p)\left[\frac{(p+p_0-\hbar k)^2}{2m}\left(1-\frac{1}{N}\right)+\frac{(p+p_0+\hbar k)^2}{2m}\frac{1}{N}\right]t/\hbar} |N,0\rangle \, .
\end{aligned}
\tag{S74}
$$

For calculating the spatial probability distribution $\langle\psi_{int}(t)|\, \hat{c}_Z^\dagger \hat{c}_Z \,|\psi_{int}(t)\rangle$, we first define the unitary operator $\hat{U}_1 = e^{-i\sum_p \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(p)\left[\frac{(p+p_0-\hbar k)^2}{2m}\left(1-\frac{1}{N}\right)+\frac{(p+p_0+\hbar k)^2}{2m}\frac{1}{N}\right]t/\hbar}$ and evaluate

$$
\begin{aligned}
&\langle\psi_{int}(t)|\, \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q) \,|\psi_{int}(t)\rangle\, e^{-i(p-q)Z/\hbar} \\
=&\frac{1}{N}\left\langle N,0 \left| \hat{U}_1^\dagger \left(\sum_l \hat{\psi}_\uparrow^\dagger(l)\hat{\psi}_\downarrow(l)\right)^\dagger \left(\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q)\right) \left(\sum_s \hat{\psi}_\uparrow^\dagger(s)\hat{\psi}_\downarrow(s)\right) \hat{U}_1 \right| N,0 \right\rangle e^{-i(p-q)Z/\hbar} \\
=&\frac{1}{N}\left\langle N,0 \left| \sum_s \hat{\psi}_\downarrow^\dagger(s)\hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q)\hat{\psi}_\downarrow(s) \right| N,0 \right\rangle e^{-i(p-q)Z/\hbar} e^{i(p-q)\left[\frac{p+q}{2m}+\left(1-\frac{1}{N}\right)\frac{p_0-k}{m}+\frac{1}{N}\frac{p_0+k}{m}\right]t/\hbar} \\
=&\left(\sum_s e^{-\frac{s^2}{2\sigma_p^2}}\right) \left\langle N-1,0 \left| \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow(q) \right| N-1,0 \right\rangle e^{-i(p-q)Z/\hbar} e^{i(p-q)\left[\frac{p+q}{2m}+\left(1-\frac{1}{N}\right)\frac{p_0-\hbar k}{m}+\frac{1}{N}\frac{p_0+\hbar k}{m}\right]t/\hbar} \\
=&(N-1)\,\phi(p)\phi(q) e^{-i(p-q)Z} e^{i(p-q)\left[\frac{p+q}{2m}+\left(1-\frac{1}{N}\right)\frac{p_0-\hbar k}{m}+\frac{1}{N}\frac{p_0+\hbar k}{m}\right]t/\hbar}.
\end{aligned}
\tag{S75}
$$

Similarly, we have

$$
\begin{aligned}
&\langle\psi_{int}(t)|\, \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q) \,|\psi_{int}(t)\rangle \\
=&\frac{1}{N}\left\langle N,0 \left| \hat{U}_1^\dagger \left(\sum_l \hat{\psi}_\uparrow^\dagger(l)\hat{\psi}_\downarrow(l)\right)^\dagger \left(\hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\uparrow(q)\right) \left(\sum_s \hat{\psi}_\uparrow^\dagger(s)\hat{\psi}_\downarrow(s)\right) \hat{U}_1 \right| N,0 \right\rangle e^{-i(p-q)Z/\hbar} \\
=&\frac{1}{N}\left\langle N,0 \left| \hat{U}_1^\dagger \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow^\dagger(q)\hat{U}_1 \right| N,0 \right\rangle e^{-i(p-q)Z/\hbar} \\
=&\frac{1}{N}\left\langle N,0 \left| \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\downarrow^\dagger(q) \right| N,0 \right\rangle e^{-i(p-q)Z/\hbar} e^{i(p-q)\left[\frac{p+q}{2m}+\left(1-\frac{1}{N}\right)\frac{p_0-\hbar k}{m}+\frac{1}{N}\frac{p_0+\hbar k}{m}\right]t/\hbar} \\
=&\phi(p)\phi(q) e^{-i(p-q)Z/\hbar} e^{i(p-q)\left[\frac{p+q}{2m}+\left(1-\frac{1}{N}\right)\frac{p_0-\hbar k}{m}+\frac{1}{N}\frac{p_0+\hbar k}{m}\right]t/\hbar}
\end{aligned}
\tag{S76}
$$

Again, since $\langle\psi_{int}(t)|\, \hat{\psi}_\downarrow^\dagger(p)\hat{\psi}_\uparrow(q) \,|\psi_{int}(t)\rangle = \langle\psi_{int}(t)|\, \hat{\psi}_\uparrow^\dagger(p)\hat{\psi}_\downarrow(q) \,|\psi_{int}(t)\rangle = 0$, we can now calculate the spatial probability distribution:

$$\langle\psi_{int}(t)|\,\hat{c}_Z^\dagger\hat{c}_Z\,|\psi_{int}(t)\rangle \propto N\exp\left[-\frac{\left(Z-\frac{\bar{p}t}{m}\right)^2}{2\sigma_Z^2(t)}\right], \tag{S77}$$

which describes a single Gaussian wave packets moving with the averaged velocity $\bar{p}/m = \frac{N-1}{N}\frac{p_0-\hbar k}{m} + \frac{1}{N}\frac{p_0+\hbar k}{m}$ as predicted by the *gedanken* experiment. We note that, unlike the mean-field calculation with spatial fringes in the wave packets, here we consider a Dicke state with no well-defined azimuthal phase on the Bloch sphere and therefore no well-defined phase of the spatial fringe.

References and Notes

1. I. D. Leroux, M. H. Schleier-Smith, V. Vuletić, Implementation of cavity squeezing of a collective atomic spin. *Phys. Rev. Lett.* **104**, 073602 (2010). doi:10.1103/PhysRevLett.104.073602 Medline

2. E. J. Davis, G. Bentsen, L. Homeier, T. Li, M. H. Schleier-Smith, Photon-Mediated Spin-Exchange Dynamics of Spin-1 Atoms. *Phys. Rev. Lett.* **122**, 010405 (2019). doi:10.1103/PhysRevLett.122.010405 Medline

3. M. A. Norcia, R. J. Lewis-Swan, J. R. K. Cline, B. Zhu, A. M. Rey, J. K. Thompson, Cavity-mediated collective spin-exchange interactions in a strontium superradiant laser. *Science* **361**, 259–262 (2018). doi:10.1126/science.aar3102 Medline

4. J. A. Muniz, D. Barberena, R. J. Lewis-Swan, D. J. Young, J. R. K. Cline, A. M. Rey, J. K. Thompson, Exploring dynamical phase transitions with cold atoms in an optical cavity. *Nature* **580**, 602–607 (2020). doi:10.1038/s41586-020-2224-x Medline

5. N. Sauerwein, F. Orsi, P. Uhrich, S. Bandyopadhyay, F. Mattiotti, T. Cantat-Moltrecht, G. Pupillo, P. Hauke, J.-P. Brantut, Engineering random spin models with atoms in a high-finesse cavity. *Nat. Phys.* **19**, 1128–1134 (2023). doi:10.1038/s41567-023-02033-3 Medline

6. A. T. Black, H. W. Chan, V. Vuletić, Observation of collective friction forces due to spatial self-organization of atoms: From Rayleigh to Bragg scattering. *Phys. Rev. Lett.* **91**, 203001 (2003). doi:10.1103/PhysRevLett.91.203001 Medline

7. K. Baumann, C. Guerlin, F. Brennecke, T. Esslinger, Dicke quantum phase transition with a superfluid gas in an optical cavity. *Nature* **464**, 1301–1306 (2010). doi:10.1038/nature09009 Medline

8. S. C. Schuster, P. Wolf, S. Ostermann, S. Slama, C. Zimmermann, Supersolid Properties of a Bose-Einstein Condensate in a Ring Resonator. *Phys. Rev. Lett.* **124**, 143602 (2020). doi:10.1103/PhysRevLett.124.143602 Medline

9. Y. Guo, R. M. Kroeze, B. P. Marsh, S. Gopalakrishnan, J. Keeling, B. L. Lev, An optical lattice with sound. *Nature* **599**, 211–215 (2021). doi:10.1038/s41586-021-03945-x Medline

10. P. Kongkhambut, J. Skulte, L. Mathey, J. G. Cosme, A. Hemmerich, H. Keßler, Observation of a continuous time crystal. *Science* **377**, 670–673 (2022). doi:10.1126/science.abo3382 Medline

11. R. M. Kroeze, Y. Guo, V. D. Vaidya, J. Keeling, B. L. Lev, Spinor Self-Ordering of a Quantum Gas in a Cavity. *Phys. Rev. Lett.* **121**, 163601 (2018). doi:10.1103/PhysRevLett.121.163601 Medline

12. G. P. Greve, C. Luo, B. Wu, J. K. Thompson, Entanglement-enhanced matter-wave interferometry in a high-finesse cavity. *Nature* **610**, 472–477 (2022). doi:10.1038/s41586-022-05197-9 Medline

13. K. C. Cox, G. P. Greve, J. M. Weiner, J. K. Thompson, Deterministic Squeezed States with Collective Measurements and Feedback. *Phys. Rev. Lett.* **116**, 093602 (2016). doi:10.1103/PhysRevLett.116.093602 Medline

14. O. Hosten, N. J. Engelsen, R. Krishnakumar, M. A. Kasevich, Measurement noise 100 times lower than the quantum-projection limit using entangled atoms. *Nature* **529**, 505–508 (2016). doi:10.1038/nature16176 Medline

15. E. Pedrozo-Peñafiel, S. Colombo, C. Shu, A. F. Adiyatullin, Z. Li, E. Mendez, B. Braverman, A. Kawasaki, D. Akamatsu, Y. Xiao, V. Vuletić, Entanglement on an optical atomic-clock transition. *Nature* **588**, 414–418 (2020). doi:10.1038/s41586-020-3006-1 Medline

16. J. M. Robinson, M. Miklos, Y. M. Tso, C. J. Kennedy, T. Bothwell, D. Kedar, J. K. Thompson, J. Ye, Direct comparison of two spin-squeezed optical clock ensembles at the 10−17 level. *Nat. Phys.* **20**, 208–213 (2024). doi:10.1038/s41567-023-02310-1

17. I. D. Leroux, M. H. Schleier-Smith, V. Vuletić, Orientation-dependent entanglement lifetime in a squeezed atomic clock. *Phys. Rev. Lett.* **104**, 250801 (2010). doi:10.1103/PhysRevLett.104.250801 Medline

18. D. Kruse, C. von Cube, C. Zimmermann, P. W. Courteille, Observation of lasing mediated by collective atomic recoil. *Phys. Rev. Lett.* **91**, 183601 (2003). doi:10.1103/PhysRevLett.91.183601 Medline

19. F. Brennecke, S. Ritter, T. Donner, T. Esslinger, Cavity optomechanics with a Bose-Einstein condensate. *Science* **322**, 235–238 (2008). doi:10.1126/science.1163218 Medline

20. M. Wolke, J. Klinner, H. Keßler, A. Hemmerich, Cavity cooling below the recoil limit. *Science* **337**, 75–78 (2012). doi:10.1126/science.1219166 Medline

21. H. Keßler, J. Klinder, M. Wolke, A. Hemmerich, Optomechanical atom-cavity interaction in the sub-recoil regime. *New J. Phys.* **16**, 053008 (2014). doi:10.1088/1367-2630/16/5/053008

22. P. Münstermann, T. Fischer, P. Maunz, P. W. H. Pinkse, G. Rempe, Observation of cavity-mediated long-range light forces between strongly coupled atoms. *Phys. Rev. Lett.* **84**, 4068–4071 (2000). doi:10.1103/PhysRevLett.84.4068 Medline

23. R. Mottl, F. Brennecke, K. Baumann, R. Landig, T. Donner, T. Esslinger, Roton-type mode softening in a quantum gas with cavity-mediated long-range interactions. *Science* **336**, 1570–1573 (2012). doi:10.1126/science.1220314 Medline

24. N. Spethmann, J. Kohler, S. Schreppler, L. Buchmann, D. M. Stamper-Kurn, Cavity-mediated coupling of mechanical oscillators limited by quantum back-action. *Nat. Phys.* **12**, 27–31 (2016). doi:10.1038/nphys3515

25. R. A. Thomas, M. Parniak, C. Østfeldt, C. B. Møller, C. Bærentsen, Y. Tsaturyan, A. Schliesser, J. Appel, E. Zeuthen, E. S. Polzik, Entanglement between distant macroscopic mechanical and spin systems. *Nat. Phys.* **17**, 228–233 (2021). doi:10.1038/s41567-020-1031-5

26. R. D. Delaney, M. D. Urmey, S. Mittal, B. M. Brubaker, J. M. Kindem, P. S. Burns, C. A. Regal, K. W. Lehnert, Superconducting-qubit readout via low-backaction electro-optic transduction. *Nature* **606**, 489–493 (2022). doi:10.1038/s41586-022-04720-2 Medline

27. E. J. Davis, A. Periwal, E. S. Cooper, G. Bentsen, S. J. Evered, K. Van Kirk, M. H. Schleier-Smith, Protecting Spin Coherence in a Tunable Heisenberg Model. *Phys. Rev. Lett.* **125**, 060402 (2020). doi:10.1103/PhysRevLett.125.060402 Medline

28. A. Shankar, L. Salvi, M. L. Chiofalo, N. Poli, M. J. Holland, Squeezed state metrology with Bragg interferometers operating in a cavity. *Quantum Sci. Technol.* **4**, 045010 (2019). doi:10.1088/2058-9565/ab455d

29. J. D. Wilson, S. B. Jäger, J. T. Reilly, A. Shankar, M. L. Chiofalo, M. J. Holland, Beyond one-axis twisting: Simultaneous spin-momentum squeezing. *Phys. Rev. A* **106**, 043711 (2022). doi:10.1103/PhysRevA.106.043711

30. J. Marino, M. Eckstein, M. S. Foster, A. M. Rey, Dynamical phase transitions in the collisionless pre-thermal states of isolated quantum systems: Theory and experiments. *Rep. Prog. Phys.* **85**, 116001 (2022). doi:10.1088/1361-6633/ac906c Medline

31. R. Richardson, N. Sherman, Exact eigenstates of the pairing-force Hamiltonian. *Nucl. Phys.* **52**, 221–238 (1964). doi:10.1016/0029-5582(64)90687-X

32. R. Richardson, N. Sherman, Pairing models of Pb206, Pb204 and Pb202. *Nucl. Phys.* **52**, 253–268 (1964). doi:10.1016/0029-5582(64)90690-X

33. F. Finger, R. Rosa-Medina, N. Reiter, P. Christodoulou, T. Donner, T. Esslinger, Spin- and Momentum-Correlated Atom Pairs Mediated by Photon Exchange and Seeded by Vacuum Fluctuations. *Phys. Rev. Lett.* **132**, 093402 (2024). doi:10.1103/PhysRevLett.132.093402 Medline

34. R. V. Pound, G. A. Rebka Jr., Apparent Weight of Photons. *Phys. Rev. Lett.* **4**, 337–341 (1960). doi:10.1103/PhysRevLett.4.337

35. T. Bothwell, C. J. Kennedy, A. Aeppli, D. Kedar, J. M. Robinson, E. Oelker, A. Staron, J. Ye, Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). doi:10.1038/s41586-021-04349-7 Medline

36. K. C. Cox, G. P. Greve, B. Wu, J. K. Thompson, Spatially homogeneous entanglement for matter-wave interferometry created with time-averaged measurements. *Phys. Rev. A* **94**, 061601 (2016). doi:10.1103/PhysRevA.94.061601

37. See the supplementary materials.

38. Z. Chen, J. G. Bohnet, J. M. Weiner, K. C. Cox, J. K. Thompson, Cavity-aided nondemolition measurements for atom counting and spin squeezing. *Phys. Rev. A* **89**, 043837 (2014). doi:10.1103/PhysRevA.89.043837

39. J. Borregaard, E. Davis, G. S. Bentsen, M. H. Schleier-Smith, A. S. Sørensen, One- and two-axis squeezing of atomic ensembles in optical cavities. *New J. Phys.* **19**, 093021 (2017). doi:10.1088/1367-2630/aa8438

40. M. A. Norcia, J. R. K. Cline, J. K. Thompson, Role of atoms in atomic gravitational-wave detectors. *Phys. Rev. A* **96**, 042118 (2017). doi:10.1103/PhysRevA.96.042118

41. A. T. Black, J. K. Thompson, V. Vuletić, On-demand superradiant conversion of atomic spin gratings into single photons with high efficiency. *Phys. Rev. Lett.* **95**, 133601 (2005). doi:10.1103/PhysRevLett.95.133601 Medline

42. R. J. Lewis-Swan, D. Barberena, J. R. K. Cline, D. J. Young, J. K. Thompson, A. M. Rey, Cavity-QED Quantum Simulator of Dynamical Phases of a Bardeen-Cooper-Schrieffer Superconductor. *Phys. Rev. Lett.* **126**, 173601 (2021). doi:10.1103/PhysRevLett.126.173601 Medline

43. D. J. Young, A. Chu, E. Y. Song, D. Barberena, D. Wellnitz, Z. Niu, V. M. Schäfer, R. J. Lewis-Swan, A. M. Rey, J. K. Thompson, Observing dynamical phases of BCS superconductors in a cavity QED simulator. *Nature* **625**, 679–684 (2024). doi:10.1038/s41586-023-06911-x Medline

44. S. Smale, P. He, B. A. Olsen, K. G. Jackson, H. Sharum, S. Trotzky, J. Marino, A. M. Rey, J. H. Thywissen, Observation of a transition between dynamical phases in a quantum degenerate Fermi gas. *Sci. Adv.* **5**, eaax1568 (2019). doi:10.1126/sciadv.aax1568 Medline

45. F. A. An, E. J. Meier, J. Ang'ong'a, B. Gadway, Correlated Dynamics in a Synthetic Lattice of Momentum States. *Phys. Rev. Lett.* **120**, 040407 (2018). doi:10.1103/PhysRevLett.120.040407 Medline

46. E. J. Meier, F. A. An, A. Dauphin, M. Maffei, P. Massignan, T. L. Hughes, B. Gadway, Observation of the topological Anderson insulator in disordered atomic wires. *Science* **362**, 929–933 (2018). doi:10.1126/science.aat3406 Medline

47. A. Celi, P. Massignan, J. Ruseckas, N. Goldman, I. B. Spielman, G. Juzeliūnas, M. Lewenstein, Synthetic gauge fields in synthetic dimensions. *Phys. Rev. Lett.* **112**, 043001 (2014). doi:10.1103/PhysRevLett.112.043001 Medline

48. T. Chalopin, T. Satoor, A. Evrard, V. Makhalov, J. Dalibard, R. Lopes, S. Nascimbene, Probing chiral edge dynamics and bulk topology of a synthetic Hall system. *Nat. Phys.* **16**, 1017–1021 (2020). doi:10.1038/s41567-020-0942-5

49. B. K. Stuhl, H. I. Lu, L. M. Aycock, D. Genkina, I. B. Spielman, Visualizing edge states with an atomic Bose gas in the quantum Hall regime. *Science* **349**, 1514–1518 (2015). doi:10.1126/science.aaa8515 Medline

50. S. Kolkowitz, S. L. Bromley, T. Bothwell, M. L. Wall, G. E. Marti, A. P. Koller, X. Zhang, A. M. Rey, J. Ye, Spin-orbit-coupled fermions in an optical lattice clock. *Nature* **542**, 66–70 (2017). doi:10.1038/nature20811 Medline

51. M. Mancini, G. Pagano, G. Cappellini, L. Livi, M. Rider, J. Catani, C. Sias, P. Zoller, M. Inguscio, M. Dalmonte, L. Fallani, Observation of chiral edge states with neutral fermions in synthetic Hall ribbons. *Science* **349**, 1510–1513 (2015). doi:10.1126/science.aaa8736 Medline

52. F. Mivehvar, H. Ritsch, F. Piazza, Cavity-Quantum-Electrodynamical Toolbox for Quantum Magnetism. *Phys. Rev. Lett.* **122**, 113603 (2019). doi:10.1103/PhysRevLett.122.113603 Medline

53. F. Mivehvar, F. Piazza, T. Donner, H. Ritsch, Cavity QED with quantum gases: New paradigms in many-body physics. *Adv. Phys.* **70**, 1–153 (2021). doi:10.1080/00018732.2021.1969727

54. T. Chanda, R. Kraus, G. Morigi, J. Zakrzewski, Self-organized topological insulator due to cavity-mediated correlated tunneling. *Quantum (Vienna)* **5**, 501 (2021). doi:10.22331/q-2021-07-13-501

55. R. Ma, M. E. Tai, P. M. Preiss, W. S. Bakr, J. Simon, M. Greiner, Photon-assisted tunneling in a biased strongly correlated Bose gas. *Phys. Rev. Lett.* **107**, 095301 (2011). doi:10.1103/PhysRevLett.107.095301 Medline

56. F. Meinert, M. J. Mark, K. Lauber, A. J. Daley, H.-C. Nägerl, Floquet Engineering of Correlated Tunneling in the Bose-Hubbard Model with Ultracold Atoms. *Phys. Rev. Lett.* **116**, 205301 (2016). doi:10.1103/PhysRevLett.116.205301 Medline

57. W. Xu, W. Morong, H.-Y. Hui, V. W. Scarola, B. DeMarco, Correlated spin-flip tunneling in a Fermi lattice gas. *Phys. Rev. A* **98**, 023623 (2018). doi:10.1103/PhysRevA.98.023623

58. B. Lücke, M. Scherer, J. Kruse, L. Pezzé, F. Deuretzbacher, P. Hyllus, O. Topic, J. Peise, W. Ertmer, J. Arlt, L. Santos, A. Smerzi, C. Klempt, Twin matter waves for interferometry beyond the classical limit. *Science* **334**, 773–776 (2011). doi:10.1126/science.1208798 Medline

59. C. Gross, Spin squeezing, entanglement and quantum metrology with Bose–Einstein condensates. *J. Phys. At. Mol. Opt. Phys.* **45**, 103001 (2012). doi:10.1088/0953-4075/45/10/103001

60. S. J. Masson, M. D. Barrett, S. Parkins, Cavity QED Engineering of Spin Dynamics and Squeezing in a Spinor Gas. *Phys. Rev. Lett.* **119**, 213601 (2017). doi:10.1103/PhysRevLett.119.213601 Medline

61. A. Periwal, E. S. Cooper, P. Kunkel, J. F. Wienand, E. J. Davis, M. Schleier-Smith, Programmable interactions and emergent geometry in an array of atom clouds. *Nature* **600**, 630–635 (2021). doi:10.1038/s41586-021-04156-0 Medline

62. C. V. Kraus, M. Dalmonte, M. A. Baranov, A. M. Läuchli, P. Zoller, Majorana edge States in atomic wires coupled by pair hopping. *Phys. Rev. Lett.* **111**, 173004 (2013). doi:10.1103/PhysRevLett.111.173004 Medline

63. W. Zheng, N. R. Cooper, Superradiance Induced Particle Flow via Dynamical Gauge Coupling. *Phys. Rev. Lett.* **117**, 175302 (2016). doi:10.1103/PhysRevLett.117.175302 Medline

64. C. Schweizer, F. Grusdt, M. Berngruber, L. Barbiero, E. Demler, N. Goldman, I. Bloch, M. Aidelsburger, Floquet approach to $\mathbb{Z}2$ lattice gauge theories with ultracold atoms in optical lattices. *Nat. Phys.* **15**, 1168–1173 (2019). doi:10.1038/s41567-019-0649-7

65. E. Colella, A. Kosior, F. Mivehvar, H. Ritsch, Open Quantum System Simulation of Faraday's Induction Law via Dynamical Instabilities. *Phys. Rev. Lett.* **128**, 070603 (2022). doi:10.1103/PhysRevLett.128.070603 Medline

66. R. Rosa-Medina, F. Ferri, F. Finger, N. Dogra, K. Kroeger, R. Lin, R. Chitra, T. Donner, T. Esslinger, Observing Dynamical Currents in a Non-Hermitian Momentum Lattice. *Phys. Rev. Lett.* **128**, 143602 (2022). doi:10.1103/PhysRevLett.128.143602 Medline

67. H. Zhang, A. Chu, C. Luo, J. K. Thompson, A. M. Rey, Control and amplification of Bloch oscillations via photon-mediated interactions. *Phys. Rev. Res.* **5**, L032039 (2023). doi:10.1103/PhysRevResearch.5.L032039

68. A. Shankar, G. P. Greve, B. Wu, J. K. Thompson, M. Holland, Continuous Real-Time Tracking of a Quantum Phase Below the Standard Quantum Limit. *Phys. Rev. Lett.* **122**, 233602 (2019). doi:10.1103/PhysRevLett.122.233602 Medline

69. C. Luo, Raw Data for Cavity-Mediated Collective Momentum-Exchange Interactions, dataset, CU Scholar (2023); https://doi.org/10.25810/6AH7-DR67.

70. A. S. Sørensen, K. Mølmer, Entangling atoms in bad cavities. *Phys. Rev. A* **66**, 022314 (2002). doi:10.1103/PhysRevA.66.022314